Czech Technical University in Prague

Faculty of Electrical Engineering
Department of Cybernetics

Classification of Journalistic Quality of Articles

Bachelor's Thesis

Olga Pimenova

Prague, May 2025

Study programme: Open Informatics
Specialization: Artificial Intelligence and Computer Science

Supervisor: Ing. Jan Šedivý, CSc.

Acknowledgments

I sincerely thank my academic supervisor, Ing. Jan Šedivý, CSc., for his expert guidance, patience, and encouragement. His profound knowledge of advanced technologies and dedication to academic excellence significantly shaped the direction and quality of my research.

I am equally grateful to my workplace supervisor, Radek Tomšů, for his practical insights and mentorship during the implementation phase. His feedback effectively bridged theoretical concepts with real-world applications, enhancing the project's outcomes.

Special thanks to the Seznam.cz recommendation systems team for providing technical resources, collaborative opportunities, and industry perspectives that strengthened the experimental aspects of this thesis.

Finally, I owe immense gratitude to my family for their unwavering emotional support, sacrifices, and belief in me. Their encouragement during challenging moments was crucial to completing this journey.

BACHELOR'S THESIS ASSIGNMENT

I. Personal and study details

Student's name: **Pimenova Olga** Personal ID number: **516221**

Faculty / Institute: **Faculty of Electrical Engineering**

Department / Institute: **Department of Cybernetics**

Study program: **Open Informatics**

Specialisation: **Artificial Intelligence and Computer Science**

II. Bachelor's thesis details

Bachelor's thesis title in English:

Classification of Journalistic Quality of Articles

Bachelor's thesis title in Czech:

Klasifikace žurnalistické kvality článků

Guidelines:

Study the issue of classifying the journalistic quality of articles using NLP technologies. Propose a classifier aimed at reducing the exposure of low-quality articles on the homepage of Seznam.cz. Implement chosen algorithms and create tests to estimate the classification accuracy. Initial data set will be provided. The main ambition is to achieve better results than the existing solution (will be provided: a simple text-based model evaluating the article's content) in order to potentially replace human editors.

Bibliography / sources:

[1] Yang, Yuting, et al. "How to write high-quality news on social network? predicting news quality by mining writing style." arXiv preprint arXiv:1902.00750 (2019).
[2] Specia, Lucia, Carolina Scarton, and Gustavo Henrique Paetzold. Quality estimation for machine translation. Springer Nature, 2022.

Name and workplace of bachelor's thesis supervisor:

Ing. Jan Šedivý, CSc. Big Data and Cloud Computing CIIRC

Name and workplace of second bachelor's thesis supervisor or consultant:

Date of bachelor's thesis assignment: **27.01.2025** Deadline for bachelor thesis submission: ___________

Assignment valid until: **20.09.2026**

prof. Dr. Ing. Jan Kybic
Head of department's signature

prof. Mgr. Petr Páta, Ph.D.
Vice-dean´s signature on behalf of the Dean

III. Assignment receipt

The student acknowledges that the bachelor's thesis is an individual work. The student must produce her thesis without the assistance of others, with the exception of provided consultations. Within the bachelor's thesis, the author must state the names of consultants and include a list of references.

Date of assignment receipt

Student's signature

FAKULTA ELEKTROTECHNICKÁ
FACULTY OF ELECTRICAL ENGINEERING
Technická 2
166 27 Praha 6

DECLARATION

I, the undersigned

Student's surname, given name(s): Pimenova Olga
Personal number: 516221
Programme name: Open Informatics

declare that I have elaborated the bachelor's thesis entitled

Classification of Journalistic Quality of Articles

independently, and have cited all information sources used in accordance with the Methodological Instruction on the Observance of Ethical Principles in the Preparation of University Theses and with the Framework Rules for the Use of Artificial Intelligence at CTU for Academic and Pedagogical Purposes in Bachelor's and Continuing Master's Programmes.

I declare that I used artificial intelligence tools during the preparation and writing of this thesis. I verified the generated content. I hereby confirm that I am aware of the fact that I am fully responsible for the contents of the thesis.

In Prague on 15.05.2025

Olga Pimenova
....
student's signature

Abstract

This thesis explores the task of automatically classifying the journalistic quality of articles in the Czech language. The motivation stems from the increasing volume of digital content, which challenges the ability of editorial teams to maintain consistent quality. To support automated assessment, a labeled dataset of Czech articles was provided by Seznam.cz, with each article annotated by human editors into one of four quality categories ranging from insufficient to high-quality.

The work involved detailed data preprocessing, including text cleaning and filtering. Significant class imbalance was observed and addressed using several techniques such as class weighting and hybrid loss functions. Various machine learning models were implemented, including transformer-based models fine-tuned for quality classification, and the CatBoost gradient boosting model trained on linguistic and structural features. Finally, a hybrid model combining both approaches was introduced.

Evaluation was conducted using multi-class classification metrics, with weighted F1 score as the primary indicator of performance. The final hybrid model showed superior results, confirming that combining deep language models with feature-based classifiers can improve the accuracy of quality prediction. The results demonstrate the potential of automated tools to support editorial decision-making and ensure content quality at scale.

Keywords natural language processing, article classification, text classification, machine learning, Czech language, journalistic quality, transformer models, CatBoost, class imbalance

Abstrakt

Tato bakalářská práce se zabývá úkolem automatické klasifikace žurnalistické kvality článků v českém jazyce. Motivací je rostoucí objem digitálního obsahu, který ztěžuje redakčním týmům udržování konzistentní kvality. Pro podporu automatizovaného hodnocení byl poskytnut anotovaný datový soubor českých článků od společnosti Seznam.cz, přičemž každý článek byl lidskými editory zařazen do jedné ze čtyř kategorií kvality – od nedostatečné až po vysokou kvalitu.

Práce zahrnovala důkladné předzpracování dat, včetně čištění a filtrování textu. Byla zjištěna výrazná nevyváženost tříd, která byla řešena pomocí několika technik, jako je vážení tříd a kombinované ztrátové funkce. Bylo implementováno několik modelů strojového učení, včetně modelů založených na transformerech, které byly doladěny pro úlohu klasifikace kvality, a modelu CatBoost, který byl trénován na lingvistických a strukturálních rysech textu. Nakonec byl navržen hybridní model, který kombinuje oba přístupy.

Vyhodnocení bylo provedeno pomocí metrik pro vícetřídní klasifikaci, přičemž hlavním ukazatelem výkonnosti byla vážená F1 míra. Konečný hybridní model dosáhl nejlepších výsledků, což potvrzuje, že kombinace hlubokých jazykových modelů s klasifikátory založenými na ručně navržených rysech může zvýšit přesnost předpovědi kvality. Výsledky ukazují potenciál automatizovaných nástrojů při podpoře redakčního rozhodování a zajištění kvality obsahu ve velkém rozsahu.

Klíčová slova zpracování přirozeného jazyka, klasifikace článků, strojové učení, čeština, kvalita článku, transformerové modely, CatBoost, nevyváženost tříd

Abbreviations

TP True Positive

FP False Positive

FN False Negative

ML Machine Learning

NLP Natural Language Processing

BERT Bidirectional Encoder Representations from Transformers

CatBoost Categorical Boosting

Insufficient Articles Class 0

Short but Sufficient Articles Class 1

Standard Articles Class 2

High-Quality Articles Class 3

BROWCzech Base-sized Roberta using Wordpiece tokenization for Czech

RetroMAE Retrieval-Oriented Masked Autoencoder

DeBERTaV3 Decoding-enhanced BERT with Disentangled Attention

ELECTRA Efficiently Learning an Encoder that Classifies Token Replacements Accurately

RTD Replaced Token Detection

MLM Masked Language Modeling

CE Cross-Entropy Loss

MSE Mean Squared Error

CatBoost Categorical Boosting

NLP Natural Language Processing

GELU Gaussian Error Linear Unit

Contents

1 Introduction

1.1 Motivation

The digital content revolution has transformed journalism into a high-velocity industry where platforms must simultaneously manage exponential content growth and maintain rigorous quality standards. Seznam.cz 2025 [1] (Fig. 1.1), serving over 5 million daily users, now employs an automated system to classify the quality of all submitted articles. To ensure the reliability of this system, a randomly selected subset of articles is reviewed manually by human editors. This approach addresses the key challenges posed by the scale and speed of modern content production:

- **Scalability limitations:** With article submissions increasing by 50% annually, manual evaluation of all content is no longer feasible
- **Consistency issues:** Human judgments can vary significantly, making automation essential for uniform quality standards
- **Quality degradation risks:** Unchecked content may harm platform credibility through misinformation or poor structure

Automated quality classification offers an effective solution by providing:

- Real-time assessment for 100% of submissions
- Consistent application of editorial guidelines
- Significant reduction in human workload
- 24/7 operational capacity without fatigue
- Data-driven insights for continuous quality improvement

Therefore, the primary motivation for this research is to develop and evaluate an automated system capable of classifying the journalistic quality of articles, replicating the nuanced decision-making of experienced human editors.

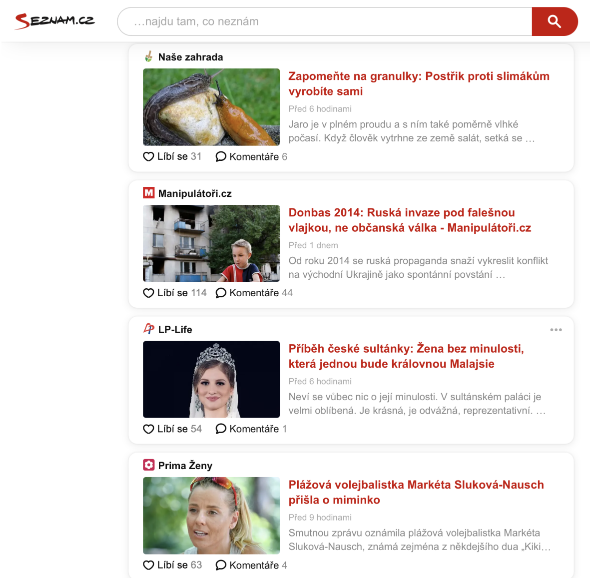

Figure 1.1: Example of Seznam.cz Homepage Displaying News Content

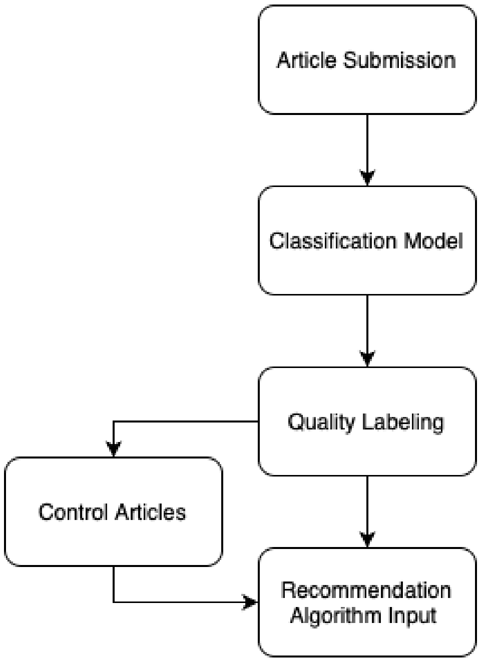

Figure 1.2: Current Editorial Workflow

1.2 Problem Formulation

The core problem addressed in this thesis is the multi-class supervised classification of journalistic articles based on their quality. Our goal is to develop a computational model that automatically assigns an article to one of four predefined quality categories, reflecting the standards used by editors at Seznam.cz.

- Class 0 (Insufficient Articles): Low-quality content that lacks structure and depth.
- Class 1 (Short but Sufficient Articles): Lacks depth and analysis, often recycling known information with limited sources.
- Class 2 (Standard Articles): Sufficiently developed content with clear structure and supported by multiple references.
- Class 3 (High-Quality Articles): Long, detailed, well-structured articles with visuals, subheadings, and references.

In the context of this thesis, **"journalistic quality"** is defined by a combination of factors that human editors consider. These include textual properties like readability, complexity, and style; structural attributes such as organization, paragraphing, and the use of headings; content characteristics like depth of analysis, factual accuracy, and source referencing; and metadata signals like the publishing channel. The aim is to create a model that learns to identify the patterns associated with these factors to predict the appropriate quality category. A significant challenge within this problem is the inherent imbalance in real-world data, where lower-quality categories are often much more frequent than the highest-quality ones.

1.3 Structure of the Thesis

This section provides an overview of the organization and content of the thesis. Chapter 1 introduces the background, motivation, and problem formulation. It also reviews related work in journalistic quality assessment.

Chapter 2 describes the data collection and analysis process. This chapter explains the sources of articles used in the study, the annotation process performed by human editors, and how the dataset is cleaned and prepared for analysis. It also reports basic statistics of the data and discusses strategies to address class imbalance.

Chapter 3 covers the evaluation metrics used to assess the classification models. It explains why accuracy is not sufficient given the class imbalance. It describes metrics such as precision, recall, and F1 score to measure model performance.

Chapter 4 outlines the methodology of the thesis. It explains the overall approach to classifying article quality, including how baseline models are developed and how advanced models are used. It also discusses feature engineering, the use of pre-trained language models and Categorical Boosting (CatBoost), and how the models are trained and tuned.

Chapter 5 presents the experiments and results. It describes the experimental setup and compares the performance of different classification models. It also shows the results of those models.

Chapter 6 provides a discussion of the findings and suggests directions for future work. It interprets the results in the context of the research goals. It also considers possible improvements or extensions.

Finally, Chapter 7 concludes the thesis by summarizing the main contributions of the work. It outlines the overall conclusions of the study.

1.4 Related Work

The automated classification of journalistic quality is a multidisciplinary challenge that integrates Natural Language Processing (NLP), Machine Learning (ML), and media studies. This section critically reviews existing literature across these fields, highlighting significant contributions and identifying gaps that this thesis addresses, particularly in the context of the Czech language media environment and platforms like Seznam.cz.

Quality Assessment Frameworks in Journalism

Traditional journalistic quality assessment relies on human evaluation guided by professional standards. Lacy and Rosenstiel 2015 [0] established eight core dimensions of journalistic quality, including accuracy, context, and sourcing diversity. While comprehensive, their framework was designed for manual evaluation, limiting its applicability to automated systems. Recent work by Calvo-Rubio and Rojas-Torrijos 2024 [3] extends these principles to AI-assisted journalism, emphasizing transparency, accountability, and ethical considerations as critical criteria for maintaining quality in digital news production. Their study highlights that AI can enhance efficiency but risks reducing nuance and context, necessitating robust evaluation frameworks.

Computational Approaches to Text Quality Assessment

Automated text quality assessment provides foundational insights for journalistic applications. Pitler and Nenkova 2008 [21] demonstrated correlations between syntactic features and human perceptions of writing quality, using metrics like sentence complexity and coherence. However, their approach was general and did not address the specific stylistic and ethical requirements of journalistic texts. More recent advancements, as noted in a 2024 article by Quantum 2024 [4], leverage NLP techniques such as grammar analysis, sentiment analysis, and text cohesion metrics to evaluate content quality automatically. These methods offer promising tools for journalism but require adaptation to domain-specific standards.

Machine Learning for Classification of Media Content

Machine learning has significantly advanced media content classification. Potthast et al. 2018 [15] developed methods to detect style-based bias in news articles, focusing on political orientation. Their work illustrates the potential of ML to identify subtle textual patterns but does not directly address overall quality. Similarly, González-Gallardo et al. 2021 [6] applied BERT-based architectures to classify news articles by credibility, achieving accuracy rates above 85%. However, their binary classification approach (credible vs. non-credible) does not capture the nuanced spectrum of journalistic quality, which often requires multi-class models.

Multi-class Classification in Natural Language Processing

Multi-class classification in NLP is a critical technique for categorizing text into multiple predefined classes, making it highly relevant for assessing the quality of journalistic content across various dimensions, such as accuracy, depth, and balance. A comprehensive review by Minaee et al. 2020 [9] examines over 150 deep learning models applied to text classification tasks, including sentiment analysis, news categorization, and question answering. The authors highlight the effectiveness of transformer-based architectures, such as BERT and its variants, which leverage large-scale pretraining and fine-tuning to achieve state-of-the-art performance in complex multi-class classification problems. These models are particularly promising for evaluating journalistic quality, as they can capture semantic nuances and contextual relationships within news articles. The review also summarizes over 40 popular datasets, some of which focus on news categorization, providing valuable benchmarks for developing quality assessment systems.

Similarly, Schmidt and Zangerle 2019 [14] introduced a multi-class approach for classifying Wikipedia article quality using document embeddings and content features. Their method, which achieved high accuracy in distinguishing quality levels, demonstrates the potential of semantic representations for quality assessment, adaptable to journalistic contexts.

Research Gaps

Despite progress, several gaps persist in the literature. First, most studies focus on English-language content, with limited exploration of languages like Czech, where morphological complexity affects NLP performance. Second, many approaches rely on binary classification, which oversimplifies the multi-dimensional nature of journalistic quality. Third, few studies address the specific needs of news aggregation platforms like Seznam.cz, where high content volume and diversity demand scalable, language-specific solutions. Finally, ethical concerns, such as bias in AI models and the loss of human judgment, remain underexplored in automated quality assessment.

This thesis aims to address these gaps by developing a multi-class classification model tailored for the Czech language media environment. By integrating textual analysis, metadata, and Czech-specific linguistic features, this work builds on existing research while tackling its limitations, particularly for news aggregation platforms.

1.5 Contributions

- A dataset of Czech articles annotated with four quality levels was prepared for multi-class classification. Texts were cleaned and filtered to ensure high-quality input data.
- Class imbalance in the dataset was mitigated by applying balancing techniques. Resampling of minority classes and class-weighting were used to improve model learning on underrepresented categories.
- Transformer-based language models, especially Decoding-enhanced BERT with Disentangled Attention (DeBERTaV3), were fine-tuned to capture semantic and contextual cues relevant to article quality.
- A CatBoost model was trained using a variety of linguistic and structural features, offering a non-transformer baseline for comparison.
- A hybrid classification model was developed by combining outputs from the transformer and CatBoost models, improving predictive performance through model fusion.
- Evaluation was based on the weighted F1 score to fairly account for class imbalance, with the hybrid model achieving the best performance among all tested approaches.
- The thesis demonstrates that automated classification of article quality is feasible and can effectively support content assessment systems in high-volume environments.

2 Data Collection and Analysis

2.1 Data Source

The data for this research originates from the internal systems of Seznam.cz. The dataset comprises articles published on the platform, along with associated metadata and quality assessments provided by the editorial team. For the purpose of this thesis, we primarily utilized the following information for each article:

- **Title:** The headline of the article.
- **Content:** The full text body of the article.
- **Channel ID:** An identifier for the specific source or section within Seznam.cz where the article was published.
- **Entity type:** Type of entity (e.g., article, video, etc.)
- **Published date:** Date when the content was published
- **Content type:** Type of content (e.g., news, blog, etc.)

The data was collected from the last 6 months.

2.2 Data Cleaning and Filtering

To prepare the dataset, the following steps were taken:

(i) Articles with fewer than 50 words were excluded to ensure sufficient content for classification.
(ii) Text was cleaned to remove emojis, HTML tags, and other non-text elements, focusing on title and main content.
(iii) Only articles were included, excluding other entity types like videos or images.

These measures ensured a clean and relevant dataset for journalistic quality classification.

2.3 Annotation Process

The quality labels used as the ground truth in this supervised learning task were assigned by experienced human editors at Seznam.cz. This annotation process is a routine part of content management on the platform, aimed at maintaining consistent editorial standards.

Editors evaluate articles based on a set of internal guidelines, but also rely significantly on their professional judgment and experience. They assess various aspects, including depth of content, writing quality, factual accuracy, structure, sourcing, and overall adherence to journalistic principles. Based on this assessment, each article is assigned one of the four quality labels defined in Section 1.2 (Insufficient Articles, Short but Sufficient Articles, Standard Articles, and High-Quality Articles).

2.4 Dataset Statistics

Dataset Size and Splits

After cleaning and filtering, the data was split into three sets: Training Set, Validation Set and Test set. The distribution of articles across the four quality labels within each dataset split is presented in Table 2.1

Table 2.1: Class Distribution Across Dataset Splits

Set	**Label 0**	**Label 1**	**Label 2**	**Label 3**	**All**
Training	9,216 (19.8%)	27,030 (58%)	9,901 (21.2%)	461 (0.9%)	46,608
Validation	2,000	5,817	2,125	109	10,051
Test	1,939	5,901	2,169	76	10,085

As clearly visible from Table 2.1, the dataset exhibits significant class imbalance. The Short but Sufficient Articles constitutes the majority (over 58%) of the articles, while the High-Quality Articles is extremely rare, representing less than 1% of the data. This imbalance presents a major challenge for model training, as standard algorithms tend to become biased towards the majority classes. Addressing this imbalance effectively is a key focus of the methodology described in the next chapter.

2.5 Data Analysis

Article length

Fig. 2.1 shows the marginal distribution of article length in words for each class. The average article length is 545 words.

The distribution is strongly right–skewed: the median is 434 words and the longest article reaches almost 9000 words. This long tail already hints at a qualitative difference between routine news briefs and in-depth reporting.

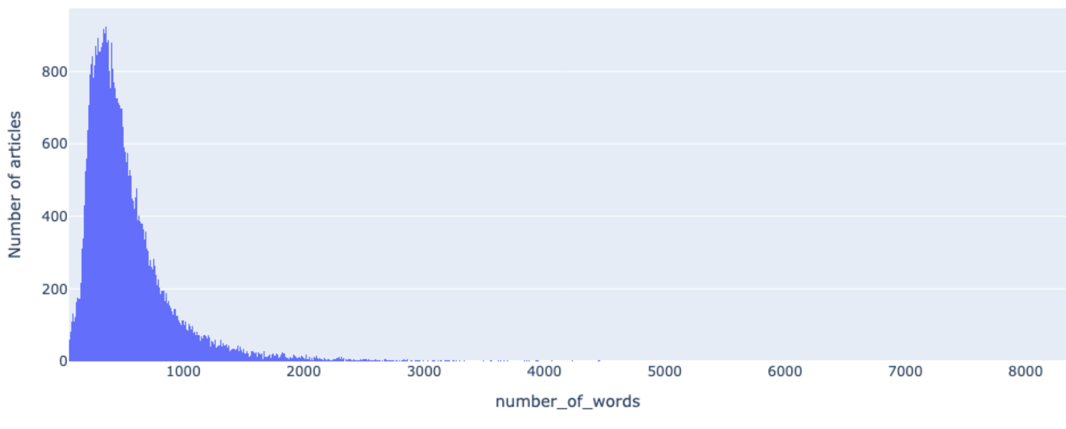

Figure 2.1: Distribution of article length in words

Length by label

Fig. 2.2 shows the distribution of article lengths for each quality label. The x-axis represents the number of words in an article, and the y-axis shows the probability density, so we can compare how article lengths differ across quality levels.

The results clearly show that better articles tend to be longer:

- Insufficient Articles and Short but Sufficient Articles articles are the shortest. Most of them are under 600 words, with label 1 forming the sharpest peak around 300-400 words.
- Standard Articles articles are longer, typically ranging from 400 to 1 200 words. They overlap with label 1 but clearly shift toward the right.
- High-Quality Articles articles are the longest. They have a wide distribution, with many texts over 1 500 words and some even exceeding 7 000 words. The long tail in the purple distribution reflects very detailed, in-depth articles.

This confirms that article length is a strong signal for quality. Longer texts are more likely to include detailed explanations, sources, and structured content—all features editors associate with high-quality journalism.

However, since the distributions overlap, article length alone is not enough to classify quality.

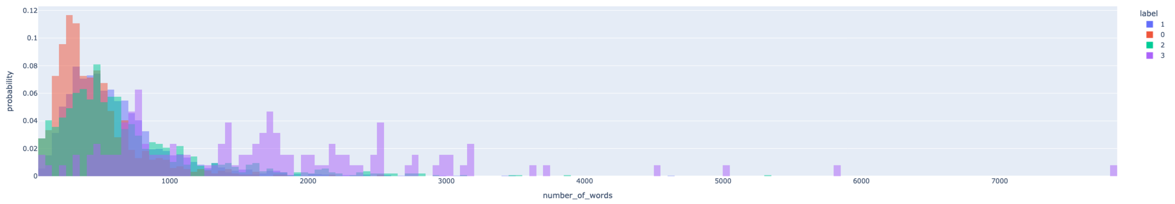

Figure 2.2: Distribution of article length in words by label

Unique-token probability density

The number of unique words in each article helps show how rich and varied the language is. Fig. 2.3 shows that higher-quality articles usually have more unique words.

Insufficient Articles often have between 280 and 380 unique words and rarely go above 800. Most articles in Short but Sufficient Articles also stay under 500 unique words. Standard Articles articles use more words, and High-Quality Articles articles often have over 1000 unique words. Some even go above 3000.

This means that better articles tend to use a wider vocabulary. However, there is still a lot of overlap between the groups, especially for shorter texts. So, the number of unique words alone cannot perfectly tell the article quality but is still a useful signal.

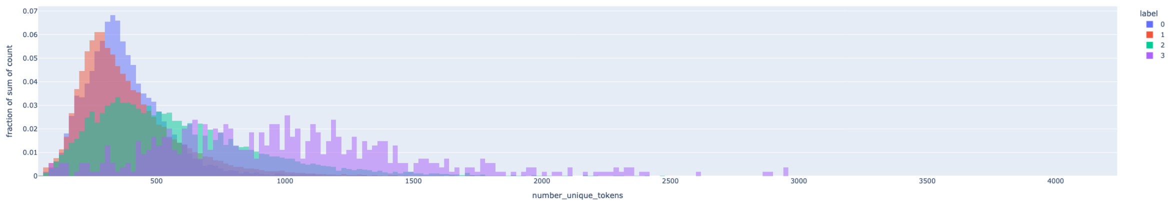

Figure 2.3: Distribution of unique tokens in articles

2.6 Dataset Balancing Strategies

Oversampling

To increase the representation of High-Quality Articles (the minority class), an attempt was made to random oversampling of High-Quality Articles instances. In this approach, existing High-Quality Articles were duplicated until their total count matched the number of Class 2 articles (around 10,000). Oversampling did improve the model's recall on High-Quality Articles examples (approximately 0.30). However, it had significant drawbacks. Training time increased by roughly 20%, and the model began to overfit the training data. Given these trade-offs, oversampling was not incorporated into the final training procedure.

Synthetic Data Generation

Another strategy was to involve generating new High-Quality Articles samples using a text-generation technique. In practice, Short but Sufficient Articles were rewritten into High-Quality Articles in an attempt to synthesize additional High-Quality Articles data. Unfortunately, the generated texts were not realistic and lacked the expected journalistic quality. The synthetic samples failed to resemble true High-Quality Articles articles, and this approach did not improve model performance. Consequently, the synthetic data generation method was abandoned.

3 Evaluation Metrics

Evaluating the performance of a multi-class classification model for journalistic article quality requires careful selection of metrics, especially given the class imbalance in the dataset. As described in the Section 2.4, the four quality categories are not equally represented. This severe imbalance means that naive models could achieve seemingly high performance by favoring the majority class. Evaluation metrics must be chosen to reflect performance across all quality levels, rather than be dominated by the majority class.

3.1 Limitations of Accuracy

Accuracy measures the proportion of correct predictions out of all predictions. While it is a simple and intuitive metric, accuracy is inadequate for imbalanced datasets. In our context of quality classification, accuracy treats all prediction errors uniformly and thus can be misleading. A model labeling every article as Short but Sufficient Articles would achieve 54% accuracy by virtue of Class 1's prevalence, potentially outperforming more nuanced models that try to distinguish all four classes but incur some mistakes on the majority class. This illustrates a key limitation: **accuracy can be artificially high even if the model entirely ignores minority classes.**

For a journalistic quality assessment system, such behavior is unacceptable – correctly identifying the minority classes (e.g. truly high-quality pieces) is crucial. Therefore, relying on accuracy alone would not highlight a model's failure to detect rare yet important cases. In light of this, we turn to alternative metrics that better capture performance across all classes, giving due weight to both common and uncommon article categories. These metrics focus on the balance between types of errors, rather than just the overall fraction correct.

3.2 Precision

Precision is calculated as:

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{3.1}$$

where True Positive (TP) is the number of correctly predicted instances of a class, and False Positive (FP) is the number of incorrectly predicted instances where the model predicted the positive class but the actual label was negative.

Precision answers the question: "*When the model predicts a certain class (especially a positive or high-quality outcome), how often is it correct?*" In the context of our highest class (High-Quality Articles), a high precision means that whenever the model flags an article as high-quality, it is usually correct. This is crucial for a platform like Seznam.cz because high precision in identifying top-quality content ensures that any article highlighted or promoted as "high-quality" truly meets the standards.

3.3 Recall

Recall measures the fraction of actual positive instances correctly identified by the model, calculated as:

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{3.2}$$

where False Negative (FN) is the number of actual positive instances that were incorrectly predicted as negative. It addresses "*How many of the actual instances of a class did the model successfully find?*" For High-Quality Articles, high recall means the model manages to identify most of the truly high-quality articles in the dataset.

3.4 F1 Score

The balance between precision and recall is a classic trade-off: improving one can often degrade the other. In our case, both are important. To illustrate:

High Precision – Ensures that articles labeled as high-quality truly meet quality standards, avoiding the risk of overestimating less suitable content.

High Recall – Ensures that most high-quality articles are identified correctly, maximizing the coverage of valuable content.

Depending on the application, one might prioritize precision over recall or vice versa. For instance, if the goal is to minimize the chance of misclassifying lower-quality content as top-tier, precision would take precedence. Alternatively, if the aim is to capture as many high-quality articles as possible, recall becomes more important—even at the cost of occasional misclassifications. In this thesis, we aim for a balance, since both types of errors carry costs.

To evaluate model performance holistically, we employ the **F1 score**, the harmonic mean of precision and recall. The F1 score yields a high value only when both precision and recall are high, making it well-suited for our imbalanced multi-class problem. It is defined as:

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{3.3}$$

This formulation captures the trade-off between FP and FN, ensuring that the model cannot achieve a high score by excelling in only one aspect.

Unlike accuracy—which treats all errors equally and can be skewed by majority classes—the F1 score ensures that minority classes are not overshadowed. A high F1 means the model is correctly identifying most high-quality articles (high recall) while avoiding mislabeling low-quality ones (high precision). F1 was used as the primary metric during model development.

3.5 Weighted F1 Score

In multi-class classification, extending precision, recall, and F1 requires aggregation across classes. For this project, the **weighted F1 score** is used as the main evaluation metric, accounting for varying class frequencies. It is calculated by computing the F1 score for each class and taking their weighted average using class support (i.e., number of instances) as weights:

$$\mathrm{F1W} = \sum_{i=1}^{n} (w_i \times \mathrm{F1}_i) \tag{3.4}$$

where $i \in \{0, 1, 2, 3\}$, $w_i = \frac{N_i}{N}$ is the fraction of samples belonging to class i in the dataset (i.e. its empirical probability) and $\mathrm{F1}_i$ is the F1 score for class i. This gives more influence to more common classes while still including all categories.

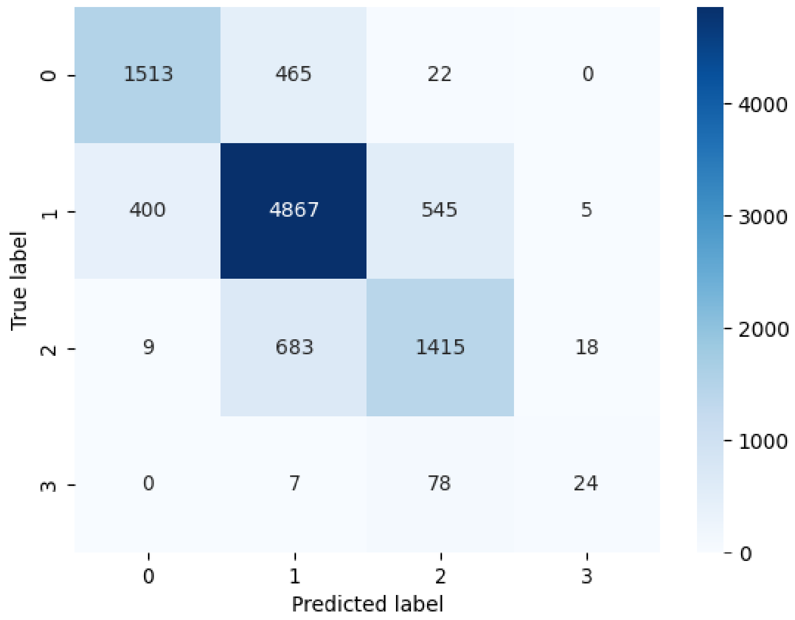

Figure 3.1: Example of Confusion Matrix. DebertaV3 Large

Why use weighted F1?

In imbalanced multi-class tasks like this one, the weighted F1 offers key benefits over macro or micro averaging:

- **Balances Class Influence:** Macro-averaged F1 weighs all classes equally, possibly overstating rare class performance. Micro-averaging, on the other hand, heavily reflects majority class behavior. Weighted F1 uses class prevalence to moderate both extremes.
- **Comprehensive Single Metric:** Weighted F1 produces a single summary that reflects performance across all classes, aiding model selection and comparison. During development, this metric was used on the validation set to guide hyperparameter tuning and model selection (4.8).
- **Reflects Real Data:** It mirrors the actual class distribution in the data, ensuring that both common and rare classes are appropriately considered without letting any one class disproportionately affect evaluation.

3.6 Confusion Matrix

The confusion matrix is a valuable visualization for understanding classification performance. It shows how instances of each true class are predicted across all classes, helping reveal systematic misclassifications. See Fig. 3.1 for an example.

In this study, the weighted F1 score was selected as the most appropriate evaluation metric and serves as the primary basis for reporting results. It was also integral to training: the final model was chosen based on its highest validation weighted F1, and early stopping was applied once this score plateaued. Using this metric consistently ensures alignment between training and evaluation phases, even under class imbalance.

3.7 Business Relevance

The choice of evaluation metrics extends beyond technical considerations; it reflects practical goals for platforms that aim to assess article quality automatically. The model is designed to approximate human editors' evaluations across a quality spectrum from Insufficient Articles to High-Quality Articles. As discussed in the Chapter 4, human-provided quality labels set a performance ceiling, and evaluation metrics should reflect how closely the model approaches that benchmark.

The weighted F1 score supports this goal by measuring model effectiveness across all quality levels, proportionally to their occurrence in the dataset. A high weighted F1 indicates that the model correctly distinguishes between different article quality levels in a way that mirrors editorial standards. This helps ensure consistency in automated quality assessment—recognizing stronger pieces and identifying weaker ones accurately, without favoring one class excessively. Ultimately, this facilitates editorial scalability while maintaining reliable quality judgments.

Cost of Classification Errors

Different misclassification types affect the system differently:

- **False Positives (FPs):** When lower-quality articles are mistakenly marked as high-quality, the perceived reliability of recommendations may decline. Users could become skeptical of the platform's ability to surface genuinely strong content.
- **False Negatives (FNs):** When high-quality articles are not correctly identified, potential value is lost, as outstanding content may fail to reach its intended audience, lowering engagement and satisfaction.

The F1 score addresses both risks by balancing precision and recall. Its weighted variant further ensures that infrequent but important categories are not neglected. By incorporating these considerations into the evaluation process, the model better supports content selection goals, ensuring that all quality levels are appropriately represented in automatic assessments.

4 Methodology

4.1 Editor Analysis and Upper Bound

Why Human Editors Are Important

When it comes to deciding what makes a good article, human judgment is key. Editors look at many factors, such as the depth of the content, the quality of the writing, the language used, and whether the article follows journalism standards. At Seznam.cz, experienced editors review articles to ensure they meet high standards. They follow clear guidelines but also rely on their own expertise. This human approach is valuable because it picks up on details and context that machines might miss. However, it's not perfect—it can be hard to stay consistent, it takes time, and it requires a lot of effort. Understanding how humans evaluate articles, including both their strengths and weaknesses, helps us design automatic systems that can do a similar job. Editors don't just focus on simple things like grammar or structure. They also check bigger issues, like whether the facts are correct, the story makes sense, and the journalism is fair and honest. But quality isn't always clear-cut—some aspects depend on personal opinions, so editors don't always agree. This is especially true for articles that aren't obviously great or terrible. These differences aren't a flaw; they just show how complex it is to judge quality.

How We Studied Editor Agreement

To understand how much editors agree and to set a target for automatic systems, we carried out a detailed study. We selected 100 articles from our dataset, choosing a mix of topics, dates, and quality levels to get a fair sample. Three experienced editors reviewed each article independently and sorted them into four categories. They followed the same rules used every day at Seznam.cz.

What We Learned About Editor Agreement

The results of our study were surprisingly strong—better than we expected. All three editors gave the same quality rating to 70 out of the 100 articles, which means a 70% full agreement rate. Even more impressive, at least two out of three editors agreed on the rating for every single article, giving us a 100% partial agreement rate. Interestingly, there was never a case where all three editors completely disagreed by picking totally different ratings. This tells us that while editors don't always see things exactly the same way, their disagreements are small—usually just between nearby categories, not about the article's overall quality. This shows a solid level of consistency among human experts.

Setting the Upper Bound

The 70% full agreement rate gives us a clear target, or "upper bound," for our automatic classification model. This is the level where even human experts fully agree, so it's a realistic goal for the system to aim for. It's also important to note that disagreements between editors were never drastic—they typically differed by just one category. This means that a reliable model should also aim to stay within this range. In practice, it's acceptable if the model's predictions differ from human ratings by at most one category, since even experienced editors make mistakes of only ± 1 level. Ensuring this small margin of error helps maintain the system's trustworthiness and mirrors real-world human judgment.

4.2 Baseline Model Performance

Before developing complex models, we established baseline classifiers to provide lower-bound reference points. Baseline models are simple heuristics that do not incorporate any deep linguistic understanding, but they set minimum performance thresholds that any useful model should exceed. We implemented two naive strategies - a Random Classifier (Section 4.2.1) and a Most Frequent Class Classifier (Section 4.2.2) - and measured their effectiveness on the dataset. We report primarily the weighted F1 score as our performance metric.

Random Classifier

The Random baseline assigns a quality label to each article by sampling from the training set's class distribution. In other words, it ignores the article text and simply guesses a label at random, but with probabilities proportional to the observed frequencies of each class in the training data. This strategy produces an expected performance near chance level. Indeed, in our dataset - which is highly imbalanced (only 0.9% of articles are High-Quality, while the majority belong to the middle categories) - the random classifier achieved a weighted F1 of approximately **0.29**. This very low F1 confirms that a classifier must do far more than random guessing to be useful. The random baseline underscores the challenge of the task: given the small fraction of high-quality articles and the dominance of the Short but Sufficient Articles, unguided guessing will mostly fail to identify the minority classes. This also highlights that any effective model needs to incorporate article-specific information (beyond class frequencies) to improve performance.

Most Frequent Class Classifier

The second baseline always predicts the single most common class from the training data for every article. In our case, the majority class was the Short but Sufficient Articles, which comprised a large portion of the dataset (Section 2.4 details the class distribution). By labeling every article as Short but Sufficient Articles, this classifier exploits the class imbalance to maximize overall accuracy, while entirely ignoring content. Perhaps surprisingly, this simplistic strategy achieved a weighted F1 of around **0.43**, significantly higher than the random baseline. The reason is that by always predicting the majority class, it correctly "classifies" all truly Short but Sufficient Article (recall for Short but Sufficient Articles is 1.0), which carry a lot of weight in the weighted F1 calculation. This result demonstrates how a skewed dataset can let a trivial predictor attain a deceptively decent aggregate score. However, the limitations of the most-frequent-class model are clear: it fails to recognize any minority class (Precision and Recall are 0 for minority classes). It cannot differentiate the subtle cues that distinguish high-quality journalism from mediocre content.

Discussion

The results of these baseline models shed light on key aspects of our classification challenge. The Random Classifier's minimal performance underscores the necessity of incorporating article-specific information to achieve meaningful outcomes, rather than depending on random predictions. In contrast, the Most Frequent Class Classifier's score of 0.43 demonstrates the pronounced effect of class imbalance, as it achieves a respectable result solely by leveraging the majority class. These baselines set clear standards for evaluating future models. Any advanced classifier—whether employing intricate feature engineering or sophisticated machine learning techniques—must exceed these initial benchmarks. Beyond improving overall metrics, we seek models that excel at correctly classifying less frequent yet significant quality categories. This establishes a critical goal for further development: to create systems that capture the nuanced elements of journalistic quality and more closely align with human editorial assessments.

4.3 Feature Engineering

The goal of the feature engineering phase is to translate each article into a set of quantitative attributes that reflect its journalistic quality. We design these features using principles from computational linguistics and journalism theory to capture various aspects of writing and content. This allows the classification model to assess quality based on interpretable characteristics of the text. (Certain implementation details are omitted here due to confidentiality, but the underlying methods and rationale are described.) The features chosen are grounded in four key dimensions of journalistic quality – **Diversity**, **Transparency**, **Attractiveness**, and **Independence** – ensuring that our approach aligns with established editorial standards and ethical AI practices. In other words, each feature group corresponds to one or more of these quality dimensions, so that the model's decisions are based on meaningful, fair criteria rather than arbitrary signals. The following subsections detail each feature group and its relation to the quality dimensions.

Linguistic Complexity Features (Attractiveness)

One important aspect of article quality is the richness and complexity of its language. Linguistic complexity features quantify the sophistication of writing, which corresponds to the Attractiveness dimension of quality – an engaging, well-crafted article typically uses varied and well-structured language. In the Czech language context, we pay special attention to syntax, vocabulary variety, and morphology. This group includes:

- **Syntactic Complexity:** Measures of sentence structure complexity, such as average sentence length and the use of subordinate clauses or compound sentences. High-quality articles often contain longer, information-rich sentences or a mix of sentence lengths that provide detail and nuance. In contrast, lower-quality pieces may rely on very short, simple sentences. We capture this by computing the average number of words per sentence and identifying complex sentence constructions. (For Czech, we note that free word order can make sentences longer or differently structured without losing clarity, so we focus on the presence of well-formed clauses and punctuation rather than word order alone.)
- **Lexical Diversity:** Metrics that assess the variety of vocabulary used, most notably the type-token ratio – the proportion of unique words (types) to total words (tokens) in the article. A higher unique word ratio indicates a broader vocabulary and more varied word choice, which is often a sign of a more informative and engaging article. As shown by the data analysis in Section 2.5, higher-quality articles tend to employ more unique words than lower-quality ones. For instance, Fig. 2.3 in Section 2.5 illustrates that High-Quality Articles often contain well over 1,000 unique tokens, whereas Insufficient Articles have far fewer unique words. This confirms that better articles use a wider vocabulary, though there is still some overlap between classes.
- **Morphological Richness:** Features that capture the range of word forms and complex words. Czech is a highly inflected language, meaning words change form (through prefixes, suffixes, case endings, etc.) and new words can be formed via derivation and compounding. We account for derived word forms and compound nouns usage in the text. A quality article might use more varied word forms (indicating precise expression and nuanced meaning) and appropriate compounds, reflecting a command of language. In summary, a high morphological variety signals that the author is leveraging the language's full expressive capability, aligning with an attractive writing style.

Together, these linguistic complexity features describe how sophisticated and varied the article's language is. They tie into the Attractiveness dimension because an article that is written with complex sentence structures, a broad vocabulary, and rich morphology is generally more engaging and indicative of higher journalistic effort. However, we also remain mindful that clarity is important – complexity should not come at the expense of understanding. (This balance is addressed by the readability features, described later.) Notably, initial data exploration showed that these features are promising: for instance, articles that editors rated as high quality exhibited significantly longer sentences and greater vocabulary variety than low-quality ones, which justifies including these metrics.

Structural Composition Features (Transparency)

Another key aspect of quality is how the article is organized and whether it adheres to good journalistic practices in its structure and use of sources. Structural composition features describe the layout and content elements of an article beyond just the raw text. These features align mostly with the Transparency dimension (and also support Independence), as they reflect how openly and clearly the article presents information and sources. High-quality journalism is usually well-structured and transparent about where information comes from. We include several features in this group:

- **Article Length:** The total length of the article (measured in words). This is a fundamental structural attribute – while not a quality guarantee on its own, length often correlates with depth of content. In Section 2.5, we observed that better-quality articles tend to be much longer on average. Fig. 2.2 in Section 2.5 shows the distribution of article lengths by quality class: High-Quality Articles have a wide length distribution with many articles over 1,500 words (and some even above 5,000 words), whereas Insufficient Articles are mostly under 600 words. This indicates that longer texts are more likely to include detailed explanations, background, and evidence – traits of high-quality reporting.
- **Section Organization:** Features capturing how the article is structured into sections or paragraphs. A well-structured news article typically has a clear introduction (lead), a body divided into coherent paragraphs or sub-sections, and sometimes a conclusion. We quantify this by looking at the number of paragraphs or explicit sections, the presence of subheadings or section titles, and the overall flow. For example, an article with many one-sentence paragraphs or a jumbled structure might indicate lower quality or a rushed piece, whereas one with logically separated sections (e.g. an introduction, several thematic paragraphs, and a wrap-up) suggests careful composition. This structural awareness contributes to transparency and readability – an article that is organized logically is easier for readers to follow and shows that the author/editors put thought into presenting the information clearly.
- **Multimedia Content Metadata:** Considers the presence and originality of multimedia elements like images, videos, and infographics. Original, journalist-taken photographs significantly enhance transparency, distinguishing higher-quality journalism from stock images. Additionally, metadata such as image descriptions or alt-text indicate editorial commitment to accessibility and clarity.
- **Citations and Source References:** Metrics evaluating how the article cites sources or includes references. A transparent article will often mention where information comes from – for example, quoting officials, referencing reports, or providing hyperlinks to source documents. We also consider source diversity in a basic way: if an article cites

several different people or organizations, it's likely providing a more balanced, independent view (related to the Independence dimension as well). This encourages the classifier to reward articles that follow the journalistic norm of attributing information, aligning with higher quality. Also we classify quotes by origin—whether sourced directly by the journalist (original quotes), from agencies, directly spoken to the channel, or indirectly through platforms like social media (Instagram). Clearly attributed original quotes signal high transparency and credibility.

Overall, the structural composition features assess whether an article is well-built in terms of format and sourcing. These features reflect the Transparency dimension because they deal with how openly the article presents its information structure and sources to the reader. A well-structured article with clear sections, supporting images, and cited sources demonstrates transparency and professionalism. These are qualities we expect in higher quality categories. Additionally, by considering structural elements, we also indirectly support the Independence dimension: for example, using multiple sources and proper citations suggests independent verification of facts rather than just copying a single source. The combination of these structural features helps the model distinguish a thoroughly crafted article from a shoddy one on more than just vocabulary – it looks at the article's journalistic form.

Readability Metrics

To assess the linguistic accessibility of journalistic articles, we implemented a suite of readability metrics tailored to the Czech language. These metrics aim to capture surface-level features that correlate with how easily a reader can comprehend a text—one key component of the "Attractiveness" dimension of journalistic quality.

While standard English readability formulas such as Flesch-Kincaid (Flesch 2007 [22]) and Gunning Fog Index (Gunning 1969 [23]) are well-known, they are not directly applicable to morphologically rich Slavic languages like Czech. Therefore, we adopted and adapted formulas that account for Czech-specific characteristics (Bendová and Cinková Aug. 2021 [5]) Our adaptation also reflects insights from previous research indicating that syllable-based complexity measures may overestimate difficulty in Slavic languages due to their inflectional nature.

Although readability is only one component of article quality, prior work suggests that high-quality journalism tends to balance clarity with intellectual richness. Thus, we hypothesized that articles in the highest-quality class would exhibit higher lexical sophistication (longer average word length, more complex sentence structures), while lower-quality articles would tend toward simplicity or inconsistency in structure. These hypotheses were borne out in later analysis, where readability features emerged as moderately informative in distinguishing between Short but Sufficient Articles and Standard Articles.

Future work may include incorporating semantic-level readability measures such as cohesion or topic coherence, which could provide a more nuanced understanding of how readers perceive clarity and engagement.

Metadata Features (Diversity)

Articles don't exist in isolation — contextual information about an article can also relate to its quality. In this context, Diversity refers to capturing a broad range of content conditions and ensuring the model is aware of different sources and patterns, so it can handle the variety present in the dataset.

Channel ID (Source Identifier): Each article in our dataset comes from a specific source or channel (for example, a particular news outlet or a section of the news platform). Initially, we included an anonymized channel identifier as a metadata feature. This allowed the model to recognize patterns in writing style or typical quality that might differ between various channels. Although this feature significantly improved classification performance, we ultimately decided not to use it in our final model.

The primary reason for excluding the channel identifier was to ensure the model classifies the articles based solely on textual content rather than relying on the reputation or historical patterns of specific channels. If we had used the channel ID, the model would strongly overfit to these identifiers—producing artificially high results by learning channel-specific biases. Consequently, if a particular channel aimed to improve its journalistic quality, the model might fail to reflect these improvements accurately, as it would still base predictions on outdated channel-specific patterns rather than on the actual article quality.

Furthermore, relying on channel identifiers could cause problems when classifying articles from channels that did not appear in the training dataset. In such cases, the model would lack critical information about these unseen channels, reducing its predictive reliability. By excluding this feature, we ensure the model remains generalizable and sensitive to the actual text content, facilitating a fair and unbiased assessment of article quality across all channels.

4.4 Performance of Pre-trained Language Models

Transformer-based language models have revolutionized NLP across multiple languages, including Czech. This chapter examines the theoretical foundations, architectures, and practical applications of pre-trained language models for Czech text classification tasks, with special attention to models specifically designed for the Czech language and their performance characteristics.

Theoretical Foundations

Pre-trained language models mark a significant advancement in NLP by shifting from traditional word-level representations to models capable of capturing nuanced contextual information. These models are built upon the transformer architecture, first introduced by Vaswani et al. 2017 [19], which relies on self-attention mechanisms to model relationships across entire sequences. The architecture, illustrated in Fig. 4.1, enables highly parallelizable computation and has become foundational for state-of-the-art models in tasks such as classification, information retrieval, and text generation.

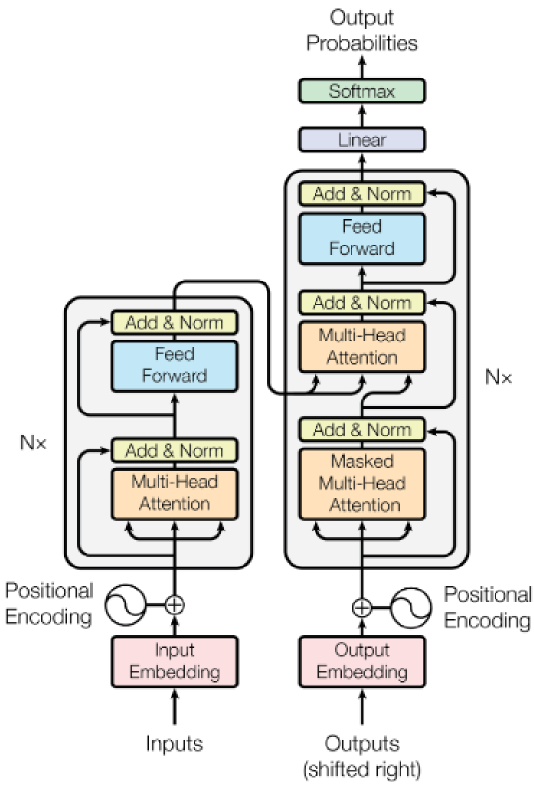

Figure 4.1: Overview of the Transformer architecture, as introduced in the paper Attention is All You Need Vaswani et al. 2017 [19]

Evaluation of Multiple Pre-trained Models

For this research, four pre-trained Bidirectional Encoder Representations from Transformers (BERT)-like (Devlin et al. 2019 [11]) language models developed internally at Seznam were evaluated. The models - Base-sized Roberta using Wordpiece tokenization for

Czech (BROWCzech), Retrieval-Oriented Masked Autoencoder (RetroMAE), DeBERTaV3 Base, and DeBERTaV3 Large - were fine-tuned to predict quality ratings for articles.

BROWCzech

Architecture BROWCzech is a monolingual Czech language model developed by Seznam.cz. It is based on the RoBERTa (Liu et al. 2019 [12]) architecture and was pre-trained on an extensive internal corpus of Czech webpages, encompassing diverse topics and writing styles. This extensive pre-training enables the model to capture the nuances of the Czech language effectively.

- **Parameters:** 109.5 million (23.4M in the embedding layer + 86.0M in the transformer layers)
- **Layers:** 12 transformer layers
- **Hidden Dimensions:** 768
- **Input Tokens:** Maximum of 512
- **Tokenizer:** Monolingual, supporting only Czech

Pre-Training The model was trained using the Masked Language Modeling (MLM) objective, where a portion of the input tokens is masked, and the model learns to predict these masked tokens. This training approach allows the model to develop a deep understanding of the language structure and semantics. The pre-training corpus consisted of a vast collection of Czech web pages, ensuring a comprehensive representation of the language used in various contexts.

Special Capabilities Due to its monolingual focus and extensive pre-training on Czech data, BROWCzech excels in tasks requiring a deep understanding of the Czech language. It is particularly effective in scenarios where the text is exclusively in Czech, providing high-quality contextual embeddings that enhance performance in downstream tasks such as text classification and sentiment analysis.

RetroMAE

Architecture RetroMAE (Bednář et al. 2024 [2]) is a transformer encoder that integrates a specialized pre-training architecture for better document-level representations. It introduces an asymmetric encoder-decoder setup during pre-trainin. The encoder is a full-scale BERT-like transformer which produces a condensed vector representation of the input text (for instance, the [CLS] token embedding). The decoder is a lightweight one-layer transformer that uses this representation to reconstruct the original text. This design does not change the model's structure at fine-tuning time - we still fine-tune the 12-layer encoder on the classification task - but the pre-training setup encourages the encoder to act as a powerful sentence/document encoder. RetroMAE uses a Czech tokenizer, so in architecture it is comparable to BROWCzech aside from the pre-training apparatus.

- **Parameters:** 130.0 million (43.9M in the embedding layer + 86.0M in the transformer layers)
- **Layers:** 12 transformer layers
- **Hidden Dimensions:** 768
- **Input Tokens:** Maximum of 512
- **Positional Embedding:** absolute
- **Tokenizer:** Monolingual, supporting only Czech

Pre-Training RetroMAE was trained on a large Czech text collection (≈253 GB) using masked autoencoding focused on retrieval. It masks 20% of tokens for the encoder and 50% for the decoder. The decoder must reconstruct the full text using the encoder output and its masked input. The encoder is large, the decoder small, forcing the encoder to learn strong semantic features. It also uses contrastive learning, bringing similar document embeddings closer and pushing different ones apart. This leads to embeddings that work well for comparing texts and help with retrieval tasks.

Special Capabilities RetroMAE captures document meaning well. It can tell if an article has deep content or is superficial. This makes it useful for judging article quality by depth and relevance. Its training also helps it detect off-topic or inconsistent content. However, it does not support other languages and may not handle foreign names well. It also shares the 512-token limit of BROWCzech. While it focuses on meaning rather than style, its embeddings may still reflect structure if it's linked to content. In general, RetroMAE is strong in identifying informative and relevant articles.

DeBERTaV3 Base

Architecture DeBERTaV3 (He, Gao, and Chen 2021 [7]) is a transformer encoder that builds on the BERT family with two major innovations: disentangled attention and improved embeddings. In DeBERTaV3, each token is characterized by two vectors - one for its content (token identity) and one for its position - instead of a single combined embeddin. During self-attention, the model can attend to content and position information separately, a mechanism known as **Disentangled Attention**. This enables a form of relative position encoding: the model doesn't just rely on fixed absolute positions, but learns relative positional biases so it can better handle long-range dependencies and varied word order. The Base model also employs an extended vocabulary (in our case, a bilingual Czech-English tokenizer) which increases the embedding layer parameters. The bilingual tokenizer and vocabulary allow it to natively process both Czech and English text. This is particularly useful for Czech news, which often contain English names, technical terms, or quotes. In summary, DeBERTaV3 Base's architecture offers a more expressive attention mechanism and cross-lingual lexical coverage, setting it apart from BROWCzech and RetroMAE which use standard attention and monolingual vocabularies.

- **Parameters:** 129.4 million (43.9M in the embedding layer + 85.5M in the transformer layers)
- **Layers:** 12 transformer layers
- **Hidden Dimensions:** 768
- **Input Tokens:** Pre-trained with sentences with max 512 tokens, generalizes up to 2048 due to relative position embeddings
- **Tokenizer:** Bilingual, supporting both Czech and English

Pre-Training DeBERTaV3 Base is trained with a method from Efficiently Learning an Encoder that Classifies Token Replacements Accurately (ELECTRA) using Replaced Token Detection (RTD) (Clark et al. 2020 [8]) instead of the usual MLM. This task teaches the model to tell real tokens from fake ones, which is more efficient and helps it learn better. It also uses Gradient-Disentangled Embedding Sharing, so embeddings can be shared without training conflicts. Pre-training was done on a Czech-English mix, helping the model understand code-switching and borrowed terms. Disentangled attention and relative position learning were also part of the training. Together, these make DeBERTaV3 Base strong in multilingual understanding and general language tasks.

Special Capabilities DeBERTaV3 Base is useful for assessing news article quality. First, it understands both Czech and English, so English quotes or names do not confuse it. This helps with the transparency of the article. Second, it captures long-distance relationships and structure, such as identifying quotes or key information placement. This supports structure and coherence assessment. Third, its training helps it spot unusual or incorrect tokens, which can help find factual mistakes. Overall, it handles diverse language, long contexts, and subtle language issues well. Its main limit is its smaller size compared to the Large version, but among base models, it is the most advanced.

DeBERTaV3 Large

Architecture The Large variant of DeBERTaV3 significantly scales up the Base model, enhancing its capacity for handling nuanced and complex language tasks. It is specifically optimized for scenarios requiring deeper semantic understanding and improved performance in demanding classification and cross-encoder tasks.

- **Parameters:** 361.4 million (58.6M in the embedding layer + 302.8M in the transformer layers)
- **Layers:** 24
- **Hidden Dimensions:** 1024
- **Input Tokens:** pre-trained with 512 tokens, generalizes up to 2048 via relative position encoding
- **Vocabulary Size:** bilingual Czech-English

Pre-Training DeBERTaV3 Large also uses RTD, but its bigger size lets it learn more details and deeper context. It was trained on a bilingual corpus like the Base version, but handles cross-lingual and complex texts better due to more parameters.

Special Capabilities Compared to the Base version, the Large model is better at tasks that need deep understanding, such as spotting small errors or complex facts. It models long texts more effectively and gives better results for article coherence and structure. This makes it ideal for high-level quality assessment and fine-grained classification tasks.

Model Selection

To compare the performance of the evaluated models, we trained them using the same loss function (described in Section 4.6), identical features, and the same context window size. Hyperparameters for each model were carefully tuned to achieve the best possible validation performance. In Table 5.1, we present the weighted F1 scores and the F1 scores for each quality class ($F1_1$–$F1_4$) on the test set.

Model	$\mathbf{F1_1}$	$\mathbf{F1_2}$	$\mathbf{F1_3}$	$\mathbf{F1_4}$	**F1 Weighted**	**Accuracy**
BROWCzech	0.727	0.752	0.656	0.359	0.711	0.70
RetroMAE	0.725	0.749	0.656	0.299	0.721	0.72
DeBERTaV3 Base	0.727	0.752	0.656	0.359	0.724	0.73
DeBERTaV3 Large	0.75	0.78	0.66	0.27	0.740	0.738

Table 4.1: Validation F1 scores for each model. $F1_1$–$F1_4$ correspond to classification quality for each label class. Accuracy is the overall accuracy of the model on the validation set.

DeBERTaV3 Large achieved the best weighted F1 score on the validation set, indicating that it is better suited for the quality classification task compared to the other models. Despite all models being trained under the same conditions, the increased model capacity and architectural improvements of DeBERTaV3 Large allowed it to generalize more effectively on unseen data.

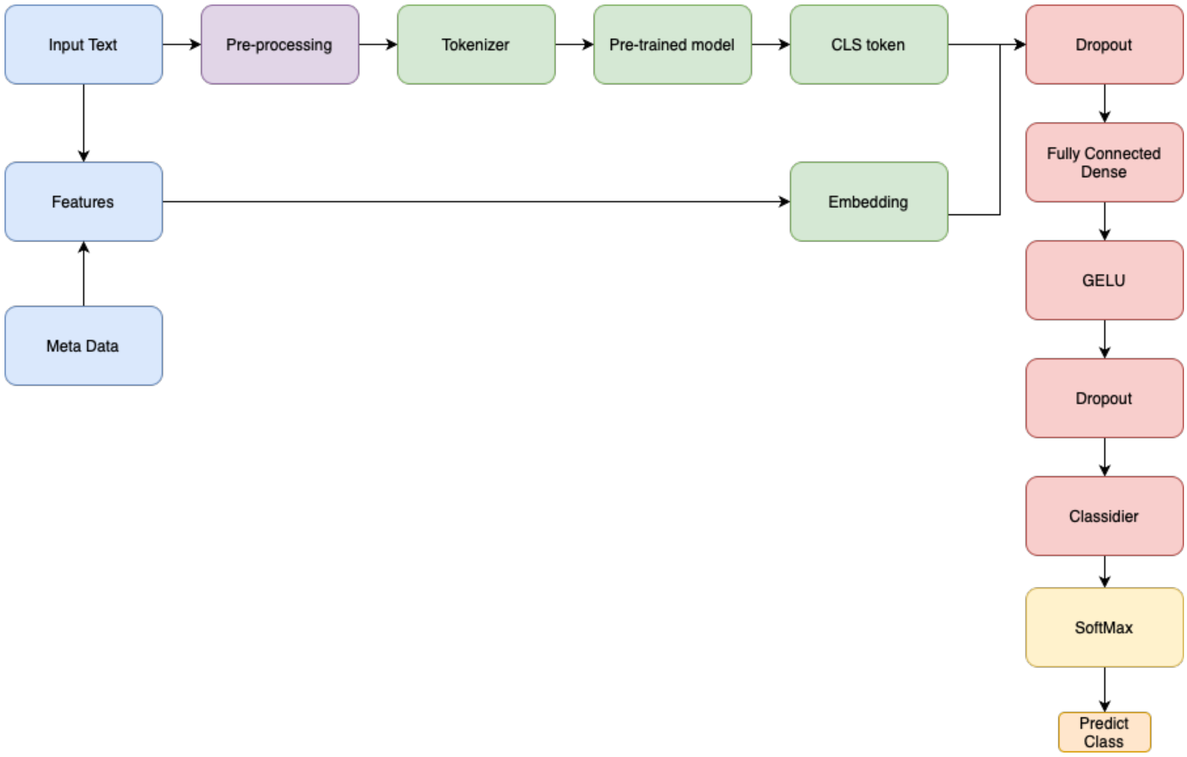

Figure 4.2: Model architecture overview for journalistic quality classification

4.5 Classifier Architecture Design

The classification architecture implemented in this research was designed to efficiently leverage the semantic information extracted by pre-trained language models while introducing minimal additional complexity to prevent overfitting, particularly given the class imbalance of the dataset.

Initially, during experiments with the BROWCzech model, a simple classification head was developed, consisting of a dropout layer, a dense projection layer, and a softmax output. However, through systematic experimentation, it became evident that a slightly deeper and regularized architecture led to improved performance without significant computational overhead. These improvements, although first tested with BROWCzech, were later adapted and finalized in conjunction with the DeBERTaV3 model, which ultimately demonstrated superior performance and became the backbone of the final system.

The overall data flow and model architecture are illustrated in Fig. 4.2. As shown, the input text is first pre-processed and tokenized, then passed through the pre-trained transformer model to obtain contextual embeddings. The [CLS] token embedding is processed through a regularized fully connected classifier head, including dropout, dense projection, Gaussian Error Linear Unit (GELU) (Hendrycks and Gimpel 2016 [20]) activation, and another dropout, before producing the final logits for prediction. Separately, additional features and metadata were extracted for possible use in ensemble models (described in Section 4.3).

4.6 Loss Function Engineering

Designing an effective loss function was critical due to the ordinal nature of the quality labels and the severe class imbalance in the dataset. The development of the loss function progressed through several stages, from a baseline categorical Cross-Entropy Loss (CE) to a final hybrid loss with an ordinal component and class weighting. Each refinement aimed to address specific shortcomings observed in earlier trials.

Initial Approach: Cross-Entropy Loss

The initial model training used the standard categorical CE loss, given its widespread success in multi-class classification. The cross-entropy loss is defined as:

$$L_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\log(p_t), \tag{4.1}$$

where N is the number of samples in batch, p_t is the predicted probability for the true class for training sample i. This loss encourages the model to assign high probability to the correct class. CE was chosen as the starting point because it generally converges faster and more reliably than regression losses for classification problems (due to its strong gradients when the predicted probability is low for the true class). In early experiments, the CE loss yielded reasonable overall accuracy. However, closer inspection revealed two major issues: it **treated all misclassifications equally**, which means that predicting a class far from the true label is penalized the same as predicting one that is nearby, and it **struggled with the extreme class imbalance in the data.**

Addressing Class Imbalance: Focal Loss vs. Class Weighting

Class imbalance is a common issue in classification tasks. Two typical strategies to mitigate imbalance are **Focal Loss** and **Class Weighting**. Focal loss dynamically scales the loss of well-classified examples, focusing training on harder, misclassified examples. In contrast, class weighting adjusts the standard CE loss by assigning higher weights to minority classes (often inversely proportional to class frequency). Each method has its own trade-offs and hyperparameters.

Focal Loss

Focal loss (introduced by Lin et al. Lin et al. 2017 [17]) modifies the standard cross-entropy as:

$$L_{\mathrm{focal}} = -\frac{1}{N}\sum_{i=1}^{N}\alpha_t(1-p_t)^{\gamma}\log(p_t),$$

where $\gamma \geq 0$ is a focusing parameter, and α_t is a weighting factor for class t. The term $(1-p_t)^{\gamma}$ down-weights easy examples and puts more emphasis on hard, misclassified ones. As a result, focal loss can help the model learn from rare or difficult examples by effectively reducing the loss contribution of well-classified, majority-class examples.

- *Dynamic scaling*: The factor $(1-p_t)^{\gamma}$ reduces the loss from easy examples ($p_t \approx 1$) and emphasizes hard examples, with γ controlling this effect.

- *Hyperparameters*: The focusing parameter γ determines how much to down-weight easy samples, while α_t can be tuned or set to class frequencies to further balance classes.
- *Use case*: Focal loss is particularly helpful when dealing with extreme class imbalance or when many easy negatives dominate the loss.

Outcomes with Focal Loss

In our setting, however, focal loss *did not yield the desired improvements*. The model trained with focal loss showed lower overall F1 than the simpler weighted loss approach, and it especially underperformed on the minority class. Several factors may explain this result:

- The imbalance in our data, while significant, is not as astronomical as in some detection problems - thus the benefit of focal modulation was less pronounced.
- Focal loss can implicitly down-weight not just easy majority-class examples but also easy examples of all classes; this means some well-classified minority instances might receive less focus than they deserve.

In practice, we found that the model still struggled to identify High-Quality Articles with focal loss - presumably because those samples were extremely rare to begin with, and focal loss alone cannot increase their presence in the training signal enough. Tuning γ did not reverse this trend, and the added complexity of focal loss did not justify itself for our problem. Similar observations have been noted in literature when using focal loss outside its original context - if the class imbalance is better handled by explicit re-weighting, focal loss can sometimes even degrade performance by over-focusing on a handful of difficult cases and under-training on the rest. Given these findings, we turned to a more direct approach: class-weighted loss, which ultimately proved more effective.

Class Weighting

Class weighting is a simpler alternative: each class k is assigned a weight w_k (often $w_k \propto \frac{1}{\text{freq}(k)}$). In the end, the chosen weights were:

- $w_0 = 1.0$ for Insufficient Articles
- $w_1 = 0.5$ for Short but Sufficient Articles
- $w_2 = 1.0$ for Standard Articles
- $w_3 = 20.0$ for High-Quality Articles

The weighted cross-entropy loss thus became

$$L^*_{CE} = -\frac{1}{N}\sum_i w_{y_i} \log p_t, \tag{4.2}$$

where w_{y_i} is the weight for the true class y_i of sample i.

- *Fixed scaling*: Weights w_k are predetermined (e.g., inversely proportional to class frequency) and remain constant during training.
- *Simplicity*: Class weighting is straightforward to implement and can often improve recall on minority classes.
- *Limitation*: If set improperly, weights can either overshoot (causing overfitting to minority classes) or undershoot (failing to mitigate imbalance effectively).

Incorporating an Ordinal Component: CE + MSE

To address the **equal-penalty problem for ordinal errors**, the loss function was augmented with a Mean Squared Error (MSE) term on the class predictions. The idea was to introduce a penalty based on the distance between the predicted class and true class, treating the class labels as numerical values on an ordinal scale. In effect, this gives a harsher penalty to predictions that are far from the true label. Such a strategy aligns with approaches in other ordinal classification domains, where a class distance penalty is added to the loss to handle ordered labels. Formally, let the classes be encoded as ordinal values $\{0, 1, 2, 3\}$ for (Insufficient, Short, Standard, High-Quality). For a given sample i, define the predicted expected class (a single continuous value) as:

$$\hat{y}_i = \sum_{c=0}^{3} c \cdot \hat{p}_{i,c}, \tag{4.3}$$

which is the expectation of the class index under the model's predicted distribution $\hat{p}_{i,c}$. The ground-truth label y_i is likewise taken as an integer in $\{0, 1, 2, 3\}$. The MSE loss is:

$$L_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2. \tag{4.4}$$

This term grows quadratically with the difference between predicted and true class indices. By summing an MSE term with the CE, the hybrid loss function encourages correct classification and additionally encourages predictions to be closer to the true class when misclassifications occur. The combined loss was formulated as:

$$L_{\text{hybrid}} = L^*_{\text{CE}} + \lambda L_{\text{MSE}}, \tag{4.5}$$

where λ controls the relative importance of the ordinal MSE term. This approach effectively blends classification and regression objectives. The expectation was that the CE component would drive the model to predict the correct class, while the MSE component would make it more sensitive to the magnitude of errors, thereby penalizing wildly incorrect predictions disproportionately.

Balancing the Loss Terms (Choosing λ)

A key step was determining an appropriate value for λ. If λ is too low, the MSE term would be negligible and the loss reduces to standard CE (failing to use the ordinal information), if λ is too high, the model might overly optimize the MSE (treating the task like regression) at the expense of classification accuracy.

To balance the contributions of the two loss components, we computed their expected values after an initial training epoch and used their ratio as a scaling factor:

$$\alpha = \frac{\mathbb{E}[L^*_{\text{CE}}]}{\mathbb{E}[L_{\text{MSE}}]}, \tag{4.6}$$

where $\mathbb{E}[L^*_{\text{CE}}]$ and $\mathbb{E}[L_{\text{MSE}}]$ denote the average CE and MSE losses, respectively. This ratio α serves as a rough estimate for selecting λ, ensuring that both terms contribute equally to the overall loss on average. We used this estimate as an initial value and then explored a range of λ values around α to fine-tune model performance.

Experiments with MSE Weight

Several models were trained with different MSE weight values to evaluate their impact. Table 4.2 summarizes the weighted F1 results for a range of λ values on the validation set:

MSE Weight λ	Weighted F1 (Validation)
0	0.69
0.5	0.70
1.0	**0.721**
2.0	0.705
3.0	0.693

Table 4.2: Performance for different MSE loss weightings for RetroMAE model

As shown, the inclusion of the MSE term yielded a noticeable improvement over using pure CE ($\lambda = 0$). Performance peaked around $\lambda \approx 1.0$, which indeed was close to the calculated α in our case, confirming that balancing the two objectives was beneficial. A small MSE contribution (0.5) already improved the model's F1, and using equal weight ($\lambda = 1.0$) for CE and MSE gave the highest F1. Increasing λ beyond 1.0 did not help further; a very large MSE weight started to slightly degrade performance (likely because the model then over-emphasized predicting the exact ordinal value, at times sacrificing classification fidelity). The best model with $\lambda = 1.0$ demonstrated that the ordinal loss component successfully reduced large misclassification errors - for example, the confusion matrix showed far fewer cases of confounding the lowest vs. highest quality classes once the MSE term was introduced, compared to the CE-only loss. Our final loss function combined this weighted CE with the MSE term:

$$L_{\text{final}} = -\frac{1}{N} \sum_{i=1}^{N} (w_{y_i} \log p_t + 1.0 \times (\hat{y}_i - y_i)^2). \tag{4.7}$$

This engineered final loss function substantially outperformed the initial approaches. It effectively balanced the model's performance across all classes. Notably, the recall for the High-Quality Articles class improved markedly. The model was able to identify a considerable portion of the High-Quality Articles that were previously being missed. This can be attributed to the 20× class weight forcing the model to pay attention to class 3, in combination with the ordinal MSE term which dissuaded the model from conflating class 3 with much lower classes.

In summary, through iterative loss function engineering, the thesis arrived at a hybrid loss that leverages the strength of CE for classification, infuses ordinal awareness via a regression term, and incorporates strategic weighting to handle class frequency skew. This nuanced loss function was pivotal in boosting the model's ability to mimic human-like judgments of article quality.

4.7 Tokenization and Sequence Handling

Each article in the dataset is first tokenized using the chosen tokenizer, converting the text into a sequence of subword tokens. The lengths of tokenized articles vary widely, so deciding on an appropriate maximum sequence length is crucial. To inform this decision, we analyzed the distribution of article lengths (in tokens) in the dataset. Fig. 4.3 shows a histogram of token counts per article, which we use to guide our sequence handling strategy.

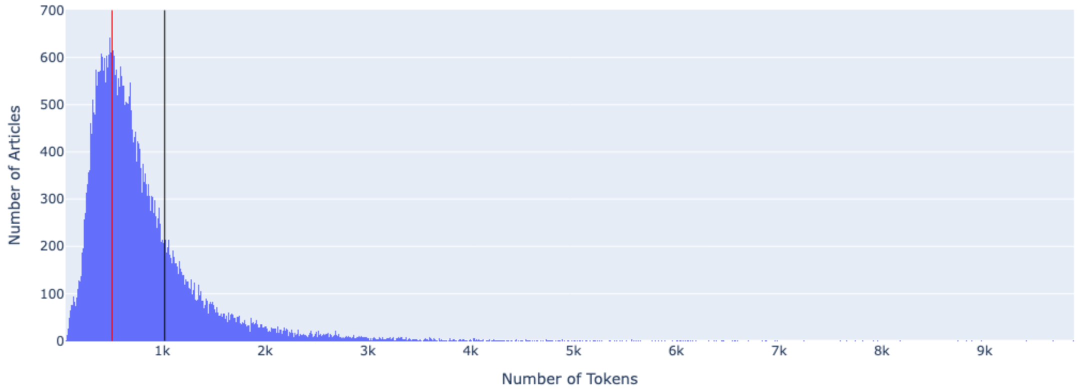

Figure 4.3: Distribution of token counts per article in the dataset. The red vertical line marks the 512-token cutoff and the black vertical line marks the 1024-token cutoff

As illustrated in Fig. 4.3, the token count distribution is strongly right-skewed. A significant portion of the articles have fewer than 512 tokens, so a *limit of 512 tokens would allow us to process most articles without truncation.* A substantial portion of articles have lengths between 512 and 1024 tokens, and beyond 1024 tokens there is a long tail of increasingly lengthy articles that are progressively rarer. In the histogram, the red vertical line marks the *512-token point* and the black vertical line *marks 1024 tokens*, corresponding to the two sequence length cutoffs we considered.

Based on this distribution, we chose an initial maximum sequence length of 512 tokens and an extended sequence length of 1024 tokens. The **512-token limit** covers the full content of most articles while keeping computational requirements manageable. Extending the **limit to 1024 tokens** allows the model to capture additional context from longer articles that exceed 512 tokens. Articles longer than 1024 tokens are relatively uncommon, so 1024 was deemed a reasonable upper bound. For any article exceeding the chosen maximum length, we apply truncation to fit the input size of the model, ensuring that we balance coverage of the article content with efficiency.

4.8 Training Methodology

After designing the hybrid loss function and addressing class imbalance, we proceeded to fine-tune the DeBERTaV3 model with a carefully planned training regime to mitigate rapid overfitting. Fine-tuning a large pre-trained model on a relatively small dataset can quickly lead to the model memorizing the training set. To counter this, our training methodology incorporated staged training, learning rate scheduling with warmup, and strong regularization through weight decay (L2 penalty) and dropout. All model training used *the final loss function* from Section 4.6.

Two-Stage Fine-Tuning Strategy

To make the most of the pre-trained DeBERTaV3 model while avoiding overfitting on limited data, we employed a structured training procedure consisting of two stages: **head-only training** followed by **full fine-tuning**.

Stage 1: Head-Only Training

- **Purpose:** Allow the classification head to learn basic decision boundaries without influencing the pre-trained backbone.
- **Method:**
  - All Transformer layers were **frozen**.
  - Only the **classifier head** (final linear layer and task-specific parameters) was trained.
- **Duration:** Training continued until the head's performance **plateaued**, typically after a few thousand steps.
- **Effect:**
  - Provided a *warm start* for the classifier.
  - Prevented early overfitting of the entire model on a small dataset.

Stage 2: Full Fine-Tuning

- **Purpose:** Fine-tune the full model to refine representations and improve classification performance.
- **Method:**
  - All DeBERTaV3 layers were **unfrozen**.
  - Training resumed using the **same optimizer** and learning rate scheduler.
  - The learning rate was **slightly reduced** to ensure stability during full-model updates.
- **Effect:**
  - Enabled the Transformer layers to adapt to task-specific patterns.
  - Resulted in a **sharp reduction in validation loss** shortly after unfreezing.

This two-stage approach allowed training to begin in a conservative regime and gradually ramp up to full capacity. By isolating the learning dynamics of the classifier head first, and only later involving the full model, we significantly reduced the risk of early overfitting while fully leveraging the representational power of DeBERTaV3.

Learning Rate Warmup and Scheduling

To prevent overfitting and training instability, we employed a tailored learning rate strategy consisting of two components: **a short warmup phase** and **cosine decay scheduling**.

Warmup Phase

- **Purpose:** Prevent sudden large weight updates early in training, when the model is still adapting to the new task.
- **Method:** Linearly increased the learning rate from a very low value to the target peak rate over the first ≈10% of training steps.
- **Effect:**
  - Avoided sharp validation loss spikes at the start of training.
  - Helped mitigate early overfitting by giving the optimizer time to find a stable direction before applying full updates.

Cosine Decay Schedule

- **Purpose:** Gradually reduce learning rate after warmup to improve convergence and stability.
- **Method:** After reaching the peak learning rate (set to 1e-5), we applied a cosine decay schedule down to a near-zero rate over the remaining training steps.
- **Effect:**
  - Enabled fast learning early in training, while reducing the risk of overshooting minima later.
  - Acted as an implicit regularizer—by the final epochs, the model could only make small, fine-grained adjustments.

Additional Notes

No learning rate restarts were used—our goal was a **smooth decay to zero**, not cyclical training.

The combination of warmup and cosine decay proved effective for stability and generalization.

Regularization and Hyperparameters

To complement our learning rate scheduling and reduce overfitting risks, we employed a combination of standard regularization techniques and carefully chosen hyperparameters.

Regularization Techniques

- **Weight Decay:** We used the AdamW optimizer (Adam with decoupled weight decay introduce by Loshchilov, Hutter, et al. 2017 [18]) with a weight decay coefficient of 1e−3 applied to all Transformer weights.
  - Acts as an L_2 regularizer by penalizing large weights.
  - Especially helpful with small datasets, where the model might otherwise overfit by driving weights to extreme values.

- **Dropout:** We retained DeBERTaV3's default dropout rate of 15% during fine-tuning.
  - Randomly zeroes activations during training to prevent reliance on specific neurons.
  - Improves generalization by introducing noise into the optimization process.

Training Setup

- **Batch size:** 16 articles.
- **Shuffling:** Training data was shuffled at each epoch to avoid learning order-specific patterns.
- **Epochs:** We trained for a maximum of 6 epochs, with evaluations:
  - At the end of each epoch.
  - Every 500 steps during training.
- We monitored both validation loss and F1 score to select the best model.

Checkpointing and Early Stopping

- Overfitting typically began after only a few epochs.
- The best model (used for final evaluation) was selected based on validation F1 and loss, occurring between 18k–20k training steps.

Training Dynamics

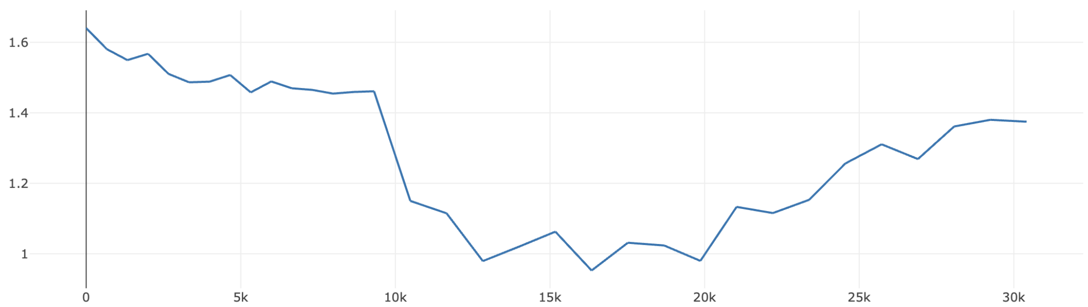

Figure 4.4: Validation loss over training steps

As shown in Figure 4.4, the validation loss illustrates key transitions in our training process:

- **Stage 1 (Head-Only Training):**
  - Validation loss declines slowly from around 1.6 to 1.4.
  - The curve is noisy and flat due to limited learning capacity in the linear head alone.
- **Stage 2 (Full Fine-Tuning):**
  - Around step 10k, a sharp drop in validation loss occurs - from ∼1.4 to ∼1.0.
  - This corresponds to unfreezing the Transformer layers, allowing deeper adaptation.
  - The minimum loss (∼0.95) is reached near step 16k.
- **Onset of Overfitting:**
  - After 20k steps, the validation loss begins to rise again.
  - By 25k–30k steps, it climbs toward 1.35+, indicating the model is memorizing training data.

Training Summary

- The curve's smooth descent and clear minimum show stable training without chaotic fluctuations.
- Our chosen checkpoint at 16k steps provided the best generalization performance.
- This confirms the value of our two-stage strategy and regularization approach.

Conclusion

Our training methodology carefully balanced optimization power and regularization:

- Warmup and staged training helped stabilize early learning.
- Weight decay and dropout reduced model complexity and overfitting risk.
- Class weighting ensured rare examples were not neglected.

These design choices enabled the model to leverage hybrid loss objectives and extract high-quality representations while maintaining generalization performance.

4.9 Model Evaluation

After adding several features from Section 4.3, the DeBERTaV3 model achieved a validation weighted F1 score of 0.766. The class-wise F1 scores were as follows:

- Class 0: F1 = 0.755
- Class 1: F1 = 0.82
- Class 2: F1 = 0.660
- Class 3: F1 = 0.20
- Accuracy: 0.77
- Weighted F1: 0.766

The confusion matrix in Fig. 4.5 shows that the model still often confuses Class 3 examples with Class 1 and Class 2. However, the overall balance of predictions has improved compared to earlier results, indicating that the hand-crafted features had a positive effect on performance.

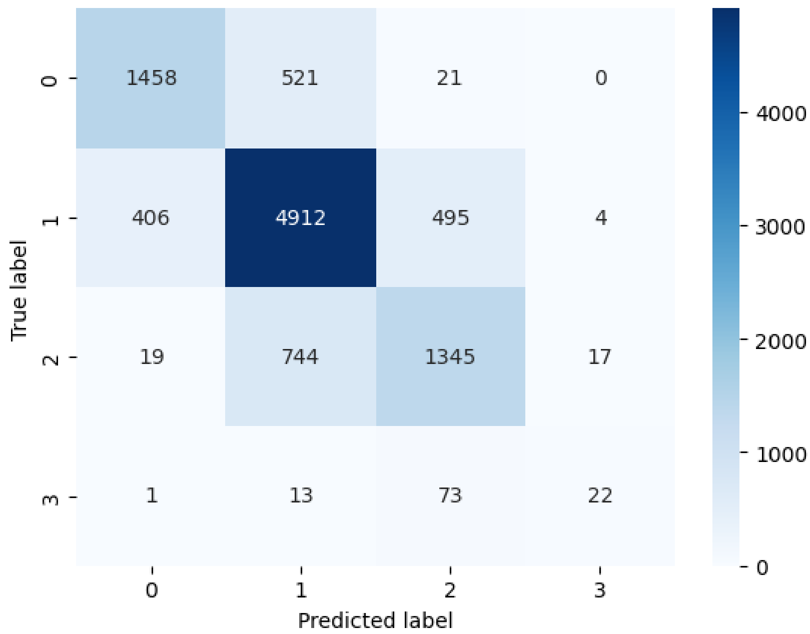

Figure 4.5: Confusion matrix of the DeBERTaV3-large + features model's predictions on the validation set.

4.10 CatBoost

We employed CatBoost introduced by Prokhorenkova et al. 2018 [16], a gradient boosting classifier, to predict article quality using the engineered feature set. CatBoost was chosen for its ability to handle categorical data natively and its robustness on imbalanced datasets. In our case, this meant we could include metadata and content-based attributes together without extensive preprocessing. For example, CatBoost can directly ingest a categorical feature like channel ID (the article's source channel) by using an ordered encoding scheme that computes target statistics from prior data points, thereby avoiding target leakage.

Another architectural advantage of CatBoost is its use of symmetric (oblivious) decision trees, where all splits at a given tree level use the same condition. This yields consistent, fast models and, coupled with the ordered boosting algorithm, helps reduce overfitting. These properties made CatBoost well-suited to our feature-based approach.

Motivation and Suitability

The motivation for using CatBoost was twofold.

First, it seamlessly handles the mix of linguistic complexity metrics, structural composition indicators, readability scores, and metadata features from Section 4.3. Unlike some algorithms, it does not require one-hot encoding or manual scaling of categorical inputs, preserving useful information encoded in features like genre or source.

Second, CatBoost's robustness to class imbalance was critical: our dataset's quality classes were highly skewed. CatBoost provides built-in options for balancing classes (e.g. an auto_class_weights = "SqrtBalanced" setting) to ensure minority classes are not ignored. This capability, combined with the algorithm's general resilience to overfitting, made it a strong candidate for our task.

Architecture and Feature Processing

We configured CatBoost to utilize all four feature groups described earlier. Internally, the model builds symmetric decision trees that split on the most informative features at each depth. The ordered boosting procedure meant that categorical features were transformed on the fly using only preceding training examples, which prevented the model from learning spurious correlations. One categorical feature, channel ID, was initially included among the inputs and, as expected, CatBoost identified it as extremely predictive of quality. This indicated that the source/publisher was a dominant signal for quality in our data. However, relying on this metadata goes against the goal of assessing content intrinsic quality. **We therefore removed channel ID in the final model**. After this adjustment, CatBoost focused on the content-based features (linguistic, structural, readability, etc.), and no single feature dominated the importance. The model could thus base its predictions on a combination of stylistic and structural characteristics of the text rather than a possibly biasing identifier.

Hyperparameter Tuning

To optimize multi-class classification performance, we performed a comprehensive search over key CatBoost hyperparameters. The tuning process focused on improving the weighted F1 score on the validation set while maintaining generalization.

Search Space and Strategy

- **Loss Function:** Compared CatBoost's standard MultiClass objective with a one-vs-all setup.
- **Boosting Iterations:** Varied the number of trees to control training duration and overfitting risk.
- **Tree Depth:** Tuned in the range of 5–10 to balance model expressiveness and regularization.
- **L2 Regularization:** Adjusted via the `l2_leaf_reg` parameter to penalize overly sharp splits.
- **Randomness Control:** Tuned the `random_strength` parameter to inject beneficial noise into the tree-splitting process and smooth decision boundaries.

Best Configuration and Observations

- The optimal setup used:
  - `MultiClass` loss function.
  - 1500 boosting iterations.
  - Tree depth of 7 to 9.
  - Small but non-zero L2 regularization.
  - Moderate `random_strength` value to improve generalization.
- **Early Stopping:** Enabled with a patience of 200 rounds.
  - Prevented overfitting by stopping training once validation performance plateaued.
  - Empirically validated our choice of 1500 trees, as performance typically stabilized before this point.

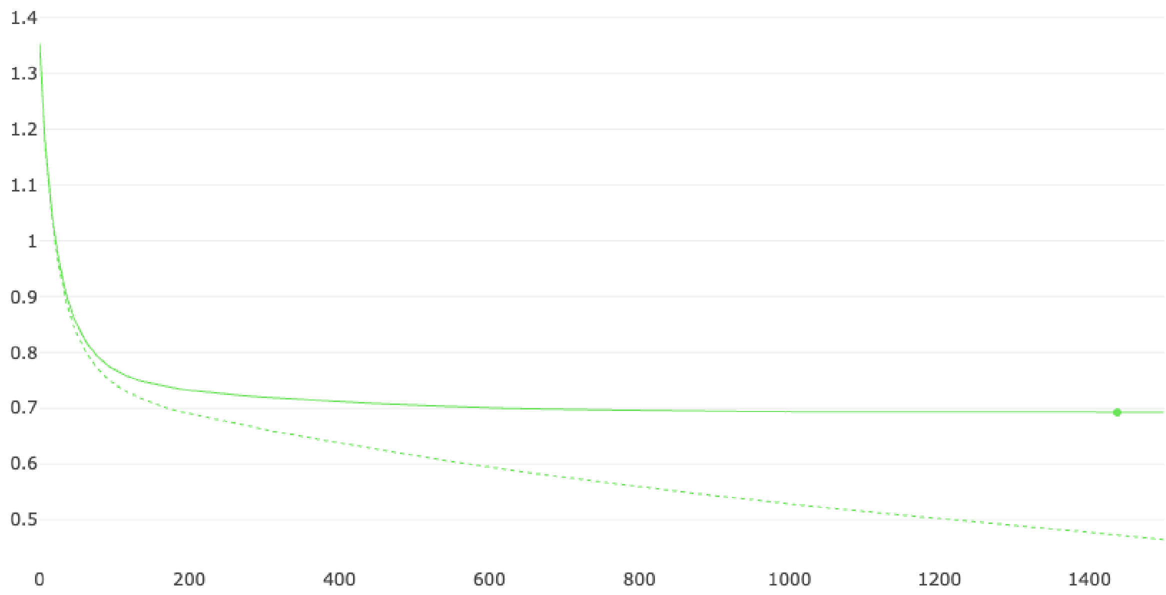

Figure 4.6: CatBoost training and validation loss over iterations

Outcome

All hyperparameters were jointly tuned using validation-based feedback. The selected configuration maximized the weighted F1 score and produced balanced class-wise performance across the multi-class prediction task.

Fig. 4.6 plots the training and validation log-loss. The curves flatten well before the stopping point, justifying the chosen iteration count.

Feature-Group Importance

To understand the contribution of each feature group to the final model, we computed the mean feature importance across all features in each group. This was done by aggregating the importance scores assigned by CatBoost to each feature and normalizing them to sum to 1. The results are shown in Table 4.3, which illustrates the relative importance of each feature group in the final model.

Group	Share of Total Importance
Structural composition	$\approx 29\,\%$
Readability metrics	$\approx 26\,\%$
Linguistic complexity	$\approx 21\,\%$
Metadata & tags	$\approx 24\,\%$

Table 4.3: Feature Group Importance Breakdown

The table shows that none of the groups dominated the model's decisions, indicating that CatBoost was **able to learn from a diverse set of features**. The structural composition and readability metrics were the most important, suggesting that the model relied heavily on these aspects to assess article quality. Linguistic complexity and metadata also contributed significantly, but to a lesser extent. This balanced feature importance distribution aligns with our expectations: high-quality articles should exhibit a combination of good structure, readability, and linguistic sophistication.

Performance and Practical Considerations

With **weighted F1=0.725** the final CatBoost model lags the neural network yet surpasses simpler baselines and remains attractive for two reasons:

1. Speed – CPU inference takes < 3 ms per article, enabling real-time scoring.

2. Interpretability – per-instance SHAP values help editors understand which textual properties influenced a given prediction.

Consequently, CatBoost serves as a complementary component in the ensemble pipeline, offering a lightweight, transparent alternative to transformer-based models.

4.11 Hybrid Model: CatBoost with Language Model Features

Based on the promising results from the standalone CatBoost model, we next experimented with a hybrid approach that **combines CatBoost with outputs from BERT-like models**. The motivation was to leverage the model's understanding of text to improve classification performance, particularly to better recognize the minority High-Quality Articles. Initial CatBoost experiments showed potential but struggled with High-Quality Articles due to its rarity, so we explored whether integrating features derived from an Pre-trained Language Models could boost the model's sensitivity to these high-quality articles. Table 4.4 summarizes the performance of the hybrid model on the validation set, comparing it to the standalone CatBoost and DeBERTaV3 models.

In the first integration attempt, we incorporated learned **DeBERTaV3 embeddings as additional features for the CatBoost** classifier. In this setup, each article's text was encoded into a dense vector representation by the DeBERTaV3, and this vector was appended to the existing feature set before training CatBoost. The intuition was that the embedding might capture complex semantic attributes of the text beyond our manually engineered features. However, **this approach did not yield any improvement** – the CatBoost model's validation metrics remained essentially unchanged when the embedding features were added. The lack of uplift suggested that either *the embedding information was redundant with our current features* or *CatBoost could not effectively leverage the high-dimensional embedding* in this configuration.

Next, we tried a different strategy: using the **DeBERTaV3's predicted class label as an input feature**. In this second experiment, the DeBERTaV3 was used as a separate classifier to predict a quality class for each article (e.g. by analyzing the text and outputting a class 0–3 label). We then fed that predicted label (as a categorical feature) into CatBoost along with all other features. The rationale was that the DeBERTaV3's prediction, even if not perfect, might cue CatBoost to certain linguistic patterns or content signals that align with quality. This method produced a **slight improvement in overall performance** – the weighted F1 score on the validation set rose to about 0.76. This was a modest gain over the baseline and indicated that the DeBERTaV3's judgment provided some useful signal. However, the improvement was not uniform across classes: High-Quality Articles in particular remained difficult to detect, with its precision and recall still very low. In other words, giving CatBoost

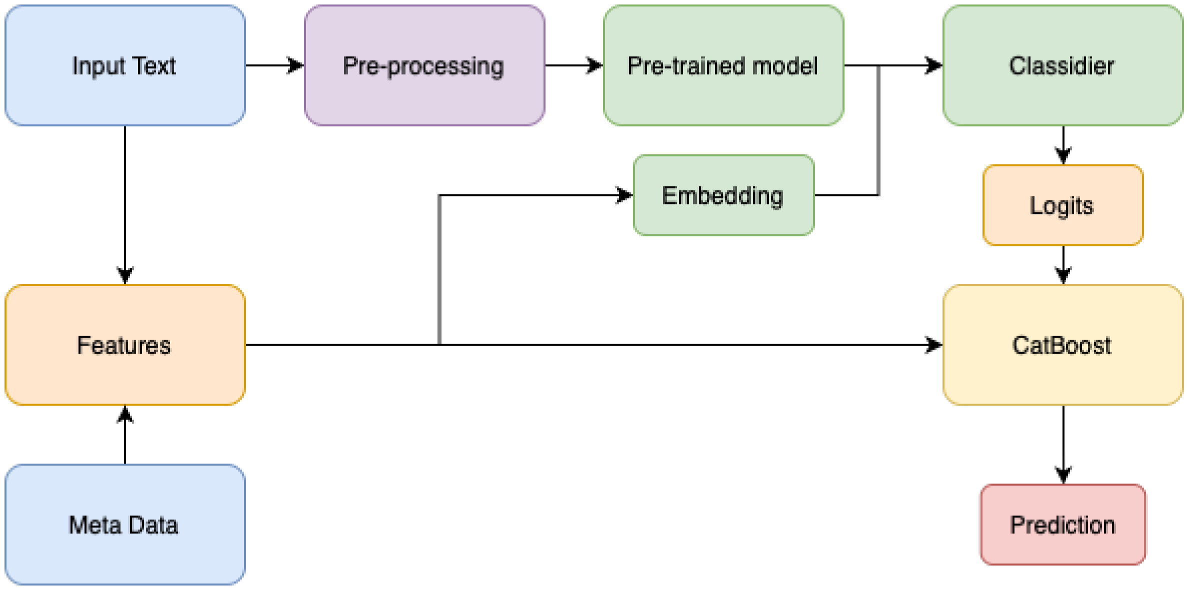

Figure 4.7: Architecture of the hybrid classification model combining DeBERTaV3 and CatBoost

the single class prediction from the DeBERTaV3 wasn't enough to substantially help with the rare high-quality articles.

Finally, our third and most successful approach was **to supply CatBoost with the full DeBERTaV3 output logits for each class**, rather than just the single best label. In this setup, the DeBERTaV3 produces a probability or confidence score for each class (0, 1, 2, 3) for a given article – essentially a vector of raw prediction scores across all classes, often referred to as logits. We included these four logit values as additional features alongside the existing engineered features and then trained the CatBoost classifier on this enriched feature set. This hybrid model formulation yielded the best results: the CatBoost + DeBERTaV3-logits model achieved a weighted F1 score of approximately 0.78 on the validation data, the highest of all tested configurations. The improvement suggests that providing the full distribution of the DeBERTaV3's predictions gave CatBoost more nuanced information. Instead of a single guess, CatBoost could learn from the DeBERTaV3's confidence levels in each class, leveraging cases where the DeBERTaV3 "hedged" between Standard Articles and High-Quality Articles, for example. This allowed the hybrid model to better differentiate articles, especially in borderline cases, and boosted overall classification performance. The overall architecture of this approach is illustrated in Fig. 4.7.

Model	**Weighted F1**	**Accuracy**
CatBoost	0.725	0.734
DeBERTaV3	0.766	0.77
CatBoost + DeBERTaV3 embedding	0.74	0.731
CatBoost + DeBERTaV3 class	0.76	0.78
CatBoost + DeBERTaV3 logits	**0.779**	**0.78**

Table 4.4: Performance of the hybrid model on the validation set.

The matrix from Fig. 4.8 shows that Class 1 (the majority class) had the highest number

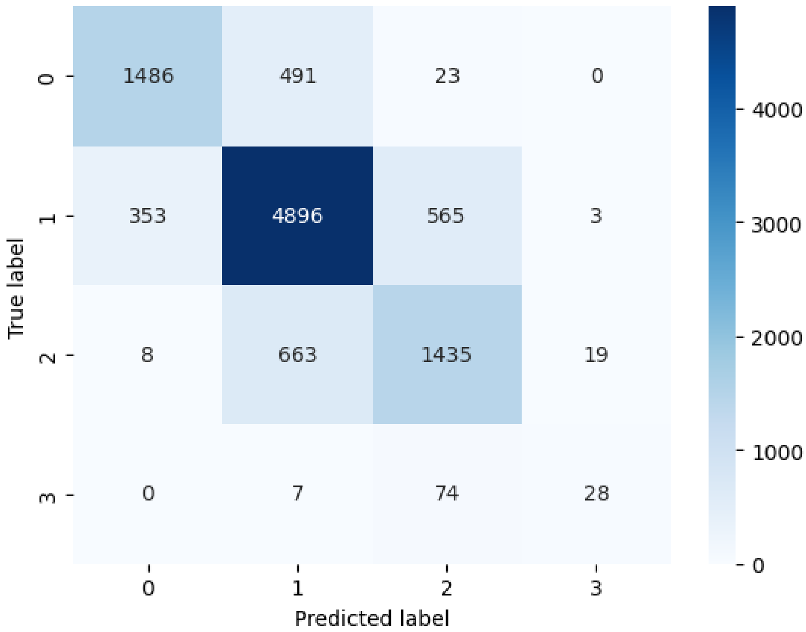

Figure 4.8: Confusion matrix of the hybrid model's predictions on the validation set

of correct predictions, with most of its errors being confusion with Class 2. In fact, the two most frequent misclassifications occurred between Class 1 and Class 2, reflecting the challenge in distinguishing adjacent quality levels. Class 3 remained the hardest category to identify correctly – the model only achieved 28 true positives for Class 3, and the majority of articles that were actually Class 3 were misclassified as Class 2. Class 0 and Class 2 also saw some confusion between each other and with Class 1, though to a lesser extent than the Class1 – Class2 overlap. This breakdown confirms that while performance for the frequent classes (0, 1, 2) was moderate to high, the High-Quality Articles category still suffered from low precision and recall due to its scarcity and the model's difficulty in separating it from Class 2.

In summary, our experiments with integrating DeBERTaV3-derived features demonstrated that embeddings or single-label predictions alone did not significantly enhance the CatBoost model. In contrast, providing the DeBERTaV3 full logits as features led to a notable improvement in classification performance. This hybrid approach successfully combined the DeBERTaV3's language understanding with CatBoost's structured learning, yielding the highest overall F1 score. Therefore, the CatBoost model enriched with DeBERTaV3 logits was selected as the final model configuration for the task, as it offered the best balance of improved accuracy (especially overall F1 ≈ 0.78) and the ability to handle all classes, including the challenging Class 3, better than previous attempts.

5 Experiments and Results

5.1 Experimental Setup

We fine-tuned each of the four pre-trained language models (BROWCzech, RetroMAE, DeBERTaV3 base, and DeBERTaV3 large) on our journalistic dataset. All models used the same data splits (described in Chapter 2). Training was performed in two stages: first we trained only the classification head for 6 epochs, then we unfroze the full model and trained all layers for 5 more epochs. This two-stage schedule (11 total epochs) was chosen to stabilize training. The models were implemented in **PyTorch** (Paszke 2019 [13]) and fine-tuned using the **Hugging Face Transformers** library (Wolf et al. 2020 [10]). We used mixed-precision (FP16) on an NVIDIA H100 PCIe GPU; the total training time for all experiments was about **27 hours**. To support experimentation, we logged metrics with **MLflow** and used **Pandas** tools to handle the data; class labels were encoded with **torchnlp LabelEncoder**.

- **Environment:** PyTorch and Hugging Face Transformers, mixed-precision training (FP16) on an NVIDIA H100 GPU (≈27 GPU-hours).
- **Tools:** MLflow, pandas/csv, torchnlp LabelEncoder.

5.2 Neural Model Results

The test performance of the models is compared in Table 5.1. The weighted F1 score of each model are shown. As expected, BROWCzech (a simpler encoder) had the lowest performance, while the DeBERTaV3 large models achieved higher accuracy. DeBERTaV3 base outperformed RetroMAE and BROWCzech, and DeBERTaV3 large improved slightly further. The best neural model was DeBERTaV3 large, with a weighted F1 around 0.766 and accuracy around 0.77.

This high-level results are consistent with our expectations. DeBERTaV3's stronger architecture led to better classification of article quality. In particular, the large model's extra capacity helped capture nuanced journalistic features. Still, all models showed difficulty with the rare High-Quality Articles. To illustrate the errors, Fig. 5.1 shows the confusion matrix for the DeBERTaV3 large model on the test data. We see that the most common class (Short but Sufficient Articles) had the most true positives, but it was sometimes confused with Standard Articles (Class 2). High-Quality Articles (Class 3) remains very hard: only a small number of high-quality articles were correctly identified, and many were misclassified as Class 1 or 2. This reflects the severe class imbalance (high-quality articles are ¡1% of the data).

5.3 CatBoost Model

We also trained a CatBoost classifier using features created in Section 4.3. The CatBoost model required only about 1 minute of training on the same GPU. This model achieved a weighted F1 around 0.716 (accuracy 72,8%). The CatBoost model was fast to train and already competitive with the neural models. However, it did not outperform the best neural model (DeBERTaV3 large). The CatBoost model's confusion matrix (Fig. 5.2) shows that it struggled with the same classes as the neural models, particularly High-Quality Articles.

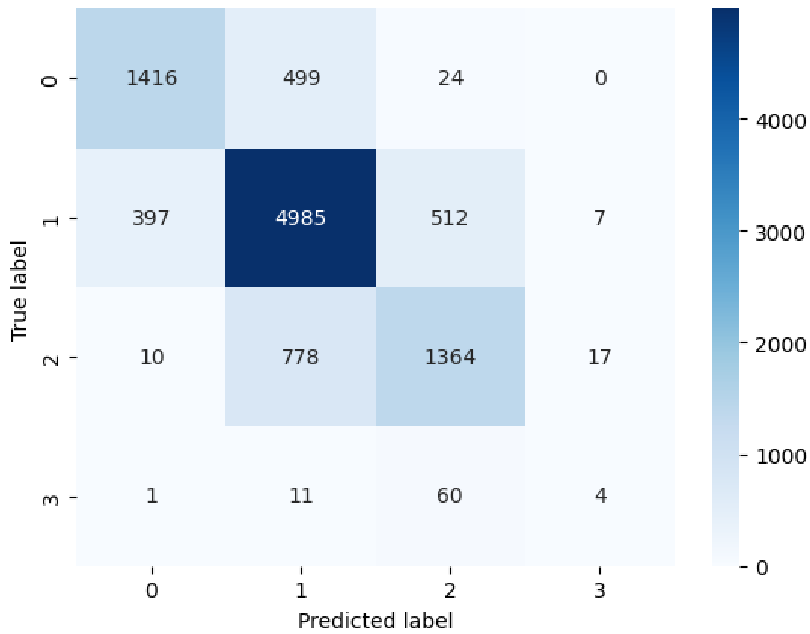

Figure 5.1: Confusion matrix for the DeBERTaV3 large model on the test set

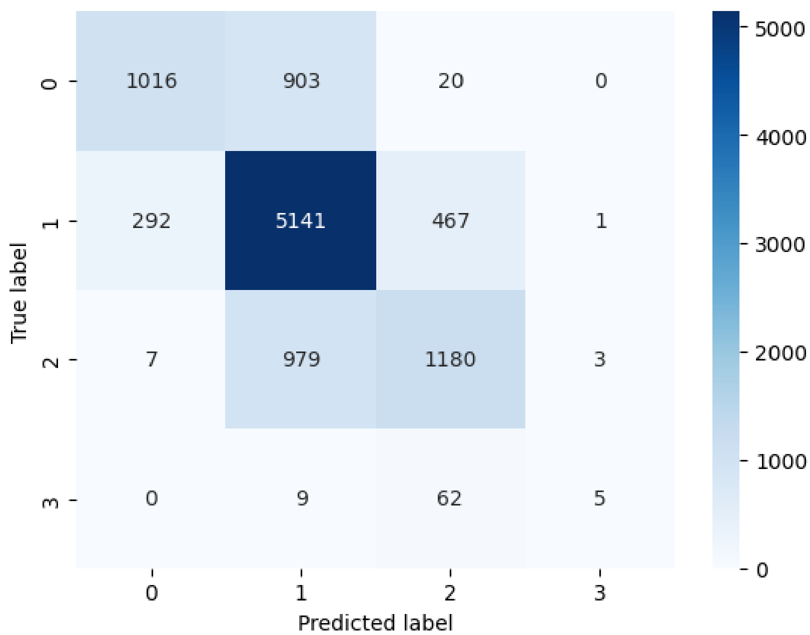

Figure 5.2: Confusion matrix for the CatBoost model on the test set

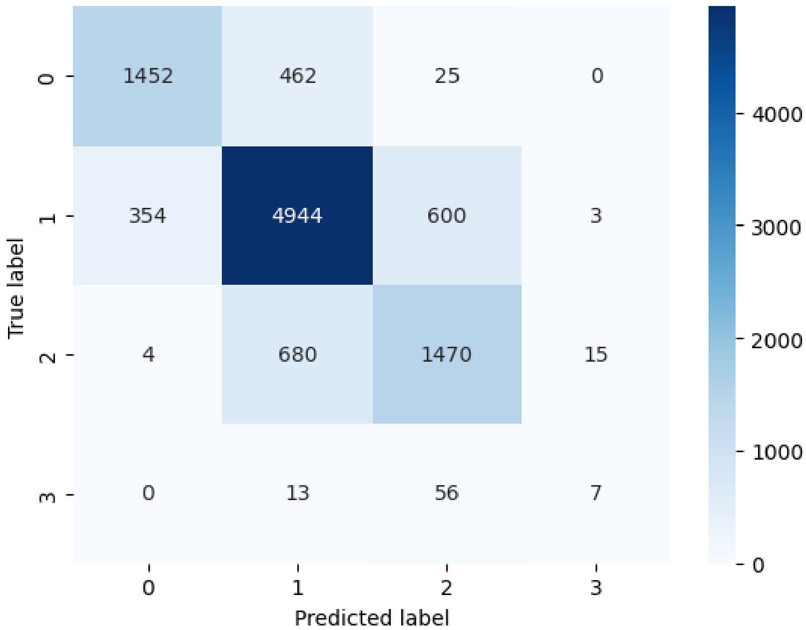

Figure 5.3: Confusion matrix for the hybrid model (DeBERTaV3 large + CatBoost) on the test set

5.4 Hybrid Model Results

The final hybrid model combined the strengths of both approaches: we fed the four logit scores from DeBERTaV3 large into the CatBoost classifier alongside other features. This hybrid (ensemble) model gave the best results of all experiments. On the test set, the hybrid model achieved a weighted F1 score of about 0.779 and an accuracy of roughly 71,1%, outperforming each individual model. These values surpass the single-model baselines and approach the human agreement upper bound (≈80%) noted in Chapter 4.1. The results demonstrate that CatBoost could effectively use the extra information in the DeBERTaV3 outputs to correct some mistakes. Fig. 5.3 shows the confusion matrix for the hybrid model on the test data. The matrix indicates improved true-positive counts in most classes. Misclassifications between Class 1 and 2 remain the largest source of error, but overall the model is more calibrated. The hybrid model's gains are especially clear in classifying borderline articles: by seeing the full DeBERTaV3 prediction vector rather than just a single guess, CatBoost learned to distinguish subtle cases better. This is evident in the confusion matrix, where fewer Standard Articles are mistaken for Short but Sufficient Articles compared to the purely neural model.

5.5 Summary of Results

In summary, the performance ranking of the models shown in the Table 5.1 were as follows: Hybrid (DeBERTaV3 large + CatBoost) > DeBERTaV3 large > DeBERTaV3 base > RetroMAE > BROWCzech > CatBoost. The hybrid model achieved the best weighted F1 (≈0.779) and accuracy (≈0.781), thanks to combining neural embeddings with gradient boosting. The CatBoost classifier alone was fast to train (≈1 minute) and already competitive, but the ensemble yielded a clear improvement. Across all models, we consistently measured weighted F1 and accuracy on the test set. The improvement from the best single model (DeBERTaV3 large) to the hybrid model is small but consistent, indicating that the additional decision-tree model helps correct a subset of errors. These results, together with the confusion matrices and charts above, show that our automated system achieves near human performance on Czech news quality classification (approaching the human agreement benchmark).

Model	**Weighted F1**	**Accuracy**
Random Classifier	0.29	0.25
Most Frequent Class Classifier	0.424	0.58
BROWCzech	0.71	0.70
RetroMAE	0.72	0.72
DeBERTaV3 base	0.724	0.73
DeBERTaV3 large	0.74	0.738
DeBERTaV3 large (features)	0.766	0.77
CatBoost (features)	0.716	0.728
Hybrid (DeBERTaV3 large + CatBoost)	**0.779**	**0.781**

Table 5.1: Test performance of the models on the validation set

6 Discussion and Future Work

6.1 Discussion

This thesis set out to build an automated system for classifying the quality of news articles. The final chosen model performed well overall, indicating that the approach is sound. In most cases, the system correctly distinguished between the predefined quality levels (Insufficient, Short but Sufficient, Standard, High-Quality). The model's predictions closely matched the categories that human editors assigned, showing that it learned to apply the editorial criteria consistently. In particular, the combination of a transformer-based language model (DeBERTaV3) and carefully engineered features proved effective in capturing important aspects of article quality, such as readability, structure, and presence of references.

The model showed several strengths. It handled the class imbalance in the data through techniques (as described in earlier chapters) so that even the rare High-Quality Articles was recognized reasonably well. The engineered features based on journalistic standards (such as measures of textual complexity, citation use, and article structure) appeared to contribute positively, as the model could detect the intended quality signals. Some confusion still remained between adjacent classes (for instance, telling apart a very good article from an excellent one), but this is also a difficulty for human editors. Overall, the results indicate that automated classification can largely mimic human judgments, although some care is needed.

6.2 Future Work

Several directions can be pursued to enhance and extend this research:

- **Integrate into the editorial workflow.** Deploy the model in the Seznam.cz production environment so that it provides real-time quality ratings for new articles. This could involve building an automated pipeline or dashboard where editors see the model's predictions. In practice, the system's output could guide editors to focus on borderline cases or flag problems early. Gathering feedback from actual editors using the tool will help improve the system.
- **Enhance explainability.** Incorporate techniques to make the model's decisions more transparent. For example, use feature-attribution methods (such as SHAP or LIME) to highlight which parts of an article influenced its quality score. Such explainability tools would allow users to understand why the model made a certain prediction, increasing trust and enabling further refinement of features.
- **Expand data and features.** Collect more data to cover a broader range of content, possibly including articles from different time periods or additional sources. Additional features could be explored, such as social engagement metrics or multi-modal signals if images are present. More data and richer inputs can improve model robustness.
- **User studies and iterative refinement.** Conduct user studies with human editors interacting with the system to identify usability issues and desired improvements. Based on this feedback, refine the model, features, and interface. This iterative development will ensure the tool meets practical editorial needs.

By pursuing these future tasks, the automated quality classification system can become more accurate, reliable, and useful. Seamless integration and better interpretability will all help bridge the gap between the model and real-world editorial use, making it a valuable assistant for maintaining high journalistic standards.

A key direction is the integration of the model into Seznam.cz's article recommendation pipeline. Incorporating the classifier's output will allow low-quality articles to be down-ranked or filtered out, directly enhancing content relevance and improving the user experience. Given that the proposed hybrid model significantly outperformed the existing channel-based heuristic it represents a strong candidate for production deployment. Its superior ability to detect low-quality articles not only improves editorial efficiency but also contributes to maintaining the platform's credibility and user trust.

7 Conclusion

The hybrid DeBERTaV3+CatBoost classifier consistently outperformed all other approaches tested in this study. On the held-out test set, it achieved the highest weighted F1 score ($\approx$0.78), exceeding the performance of individual models and approaching the estimated human inter-annotator agreement. In practical terms, this means our system's quality judgments are nearly on par with those of experienced human editors. Moreover, by accurately flagging low-quality articles, the model offers direct practical benefits: it can be used to demote or filter out poor-quality content in the recommendation and curation pipeline, thereby improving the overall quality of material presented to users. In summary, the proposed model not only achieved the best results of all models tested, but also meets the goal of human-like performance in automated article quality assessment.

8 References

[1] Seznam.cz. *Seznam.cz Homepage*. https://www.seznam.cz/. Accessed: 2025-05-05. 2025.

[2] Jiří Bednář et al. "Some like it small: Czech semantic embedding models for industry applications". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 38. 21. 2024, pp. 22734–22742.

[3] Luis M. Calvo-Rubio and José L. Rojas-Torrijos. "Criteria for journalistic quality in the use of artificial intelligence". In: *Communication & Society* 37.2 (2024), pp. 247–259. DOI: 10.15581/003.37.2.247-259. URL: https://www.researchgate.net/publication/380169433_Criteria_for_journalistic_quality_in_the_use_of_artificial_intelligence.

[4] Sophia Quantum. *Automated content quality assessment with AI*. https://medium.com/@SophiaQuantum/automated-content-quality-assessment-with-ai-6719b326ab15. Accessed: January 2025. 2024.

[5] Klára Bendová and Silvie Cinková. "Adaptation of Classic Readability Metrics to Czech". In: Aug. 2021, pp. 159–171. ISBN: 978-3-030-83526-2. DOI: 10.1007/978-3-030-83527-9_14.

[6] Carlos González-Gallardo et al. "Stance detection with BERT embeddings for credibility analysis". In: *Procesamiento del Lenguaje Natural* 67 (2021), pp. 169–180. URL: https://asu.elsevierpure.com/en/publications/stance-detection-with-bert-embeddings-for-credibility-analysis-of.

[7] Pengcheng He, Jianfeng Gao, and Weizhu Chen. "Debertav3: Improving deberta using electra-style pre-training with gradient-disentangled embedding sharing". In: *arXiv preprint arXiv:2111.09543* (2021).

[8] Kevin Clark et al. "Electra: Pre-training text encoders as discriminators rather than generators". In: *arXiv preprint arXiv:2003.10555* (2020).

[9] Shervin Minaee et al. "Deep Learning Based Text Classification: A Comprehensive Review". In: *arXiv preprint arXiv:2004.03705* (2020). URL: https://arxiv.org/abs/2004.03705.

[10] Thomas Wolf et al. "Transformers: State-of-the-art natural language processing". In: *Proceedings of the 2020 conference on empirical methods in natural language processing: system demonstrations*. 2020, pp. 38–45.

[11] Jacob Devlin et al. "Bert: Pre-training of deep bidirectional transformers for language understanding". In: *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 2019, pp. 4171–4186.

[12] Yinhan Liu et al. "Roberta: A robustly optimized bert pretraining approach". In: *arXiv preprint arXiv:1907.11692* (2019).

[13] A Paszke. “Pytorch: An imperative style, high-performance deep learning library”. In: *arXiv preprint arXiv:1912.01703* (2019).

[14] Miriam Schmidt and Eva Zangerle. “Article Quality Classification on Wikipedia: Introducing Document Embeddings and Content Features”. In: *Proceedings of the 15th International Symposium on Open Collaboration, OpenSym ’19*. New York, NY, USA: Association for Computing Machinery, 2019. ISBN: 9781450363198. URL: https://dl.acm.org/doi/10.1145/3306446.3340831.

[15] Martin Potthast et al. “A stylometric inquiry into hyperpartisan and fake news”. In: *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*. 2018, pp. 231–238. URL: https://ojs.aaai.org/index.php/AAAI/article/view/5386/5242.

[16] Liudmila Prokhorenkova et al. “CatBoost: unbiased boosting with categorical features”. In: *Advances in neural information processing systems* 31 (2018).

[17] Tsung-Yi Lin et al. “Focal loss for dense object detection”. In: *Proceedings of the IEEE international conference on computer vision*. 2017, pp. 2980–2988.

[18] Ilya Loshchilov, Frank Hutter, et al. “Fixing weight decay regularization in adam”. In: *arXiv preprint arXiv:1711.05101* 5 (2017), p. 5.

[19] Ashish Vaswani et al. “Attention is all you need”. In: *Advances in neural information processing systems* 30 (2017).

[20] Dan Hendrycks and Kevin Gimpel. “Gaussian error linear units (gelus)”. In: *arXiv preprint arXiv:1606.08415* (2016).

[0] Stephen Lacy and Tom Rosenstiel. *Defining and Measuring Quality Journalism*. https://www.jwac.org/Lacy.pdf. Accessed: January 2025. 2015.

[21] Emily Pitler and Ani Nenkova. “Revisiting readability: A unified framework for predicting text quality”. In: *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*. 2008, pp. 186–195. URL: https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=a80837b2bbe847aa9ce82ea9a14d23d82b4e4119.

[22] Rudolf Flesch. “Flesch-Kincaid readability test”. In: *Retrieved October* 26.3 (2007), p. 2007.

[23] Robert Gunning. “The fog index after twenty years”. In: *Journal of Business Communication* 6.2 (1969), pp. 3–13.